Rotational Diagonal Kalman Linear Attention: Model, Lessons from Mamba-2/Mamba-3, and Production Implementation

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Abstract

Mamba-2 supplies a strong selective state-space baseline, but its SISO state transition is a scalar radial contraction: it cannot rotate a two-dimensional latent subspace. The installed Mamba-3 SISO implementation adds rotation economically by accumulating a bounded angular rate and applying the same rotary transform to queries and keys. Its most useful lesson is therefore not a particular state-space write rule, but a representation: pairwise rotation can be moved out of the recurrent state update and into a phase-rotated query/key basis.

Rotational diagonal Kalman linear attention (RDiag KLA, E2 clock) adopts that representation while retaining KLA's diagonal Kalman covariance and innovation write. It pair-ties radial transition and process-noise parameters, uses a detached live radial statistic to set a bounded token/head clock, scans phase in a documented negative gauge, and isolates the phase VJP from radial dynamics. This note defines the model, the precise finite-precision implementation contract, and the tests used for promotion. The sealed E2 baseline implementation is commit `c3346feaecae1486cdbc71f82b4edeacc4198950`; its immutable production snapshot passed the complete one-H200 correctness and performance gate (117 core RDiag tests plus 54 E2 tests). Distributed training-system checks and the multi-node soak/attestation remain before the planned 750M DiagKLA production submission. The one-H200 result concerns implementation correctness and measured kernel/layer performance, not the eventual language-model quality of a long training run. A newer shared-DT investigation, including its unresolved production-qualification status, is reported in Section 10.

1 Scope and notation

Let B, T, H, K, V, D denote batch size, sequence length, number of heads, key dimension per head, value dimension per head, and model width. The rotary prefix has even width $K_r = 2P$, where P is its number of adjacent pairs. Tensors use the following shapes:

$$\begin{aligned} &\boldsymbol{x} \in \mathbb{R}^{B\times T\times D}, && \boldsymbol{q}, \boldsymbol{k}, \boldsymbol{\alpha}, \boldsymbol{\omega} \in \mathbb{R}^{B\times T\times H\times K},\\ &\boldsymbol{v} \in \mathbb{R}^{B\times T\times H\times V}, && \boldsymbol{e} \in \mathbb{R}^{B\times T\times P},\\ &\boldsymbol{\tau} \in \mathbb{R}^{B\times T\times H}, && \boldsymbol{\delta}, \boldsymbol{\phi} \in \mathbb{R}^{B\times T\times H\times P}. \end{aligned}$$

The raw angle e_{btj} is shared across heads; the clock τ_{bth} makes the resulting increment head-specific. For an adjacent pair $(2j, 2j+1)$, define

$$R(\vartheta) = \begin{bmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{bmatrix}, \qquad R(\vartheta)^\mathsf{T} R(\psi) = R(\psi - \vartheta). \tag{1}$$

All equations below are per batch element unless a batch index is useful. “SISO” follows the Mamba implementation’s terminology; a head can still carry multiple value channels that share the same scalar state-space coefficients.

2 What Mamba-2 has—and what it lacks

For head h, value channel p, and SSM state coordinate n, the Mamba-2 selective SISO update can be written

$$d_{th} = \operatorname{softplus}(\widetilde{d}_{th} + b^d_h), \qquad a_{th} = \exp(A_h d_{th}), \qquad A_h < 0, \tag{2}$$

$$X_{t,h,p,n} = a_{th} X_{t-1,h,p,n} + d_{th} B_{t,n} u_{t,h,p}, \qquad y_{t,h,p} = \sum_n C_{t,n} X_{t,h,p,n} + D_h u_{t,h,p}. \tag{3}$$

The input projection is ordered as $[z, x, B, C, dt]$. It emits one dt row per token and head; B and C are token-dependent state vectors. The factor a_{th} is a scalar radial contraction over the head state, and $d_{th} B_t u_t$ is its discretized injection. There is no angle projection, phase state, adjacent-pair structure, sine/cosine operation, or 2×2 rotation in this path. Mamba-2 therefore supplies selective decay and injection, but not oscillatory transport. A pair of real coordinates within a head shares the same scalar decay, but that factor is $a_{th} I$: it has no skew component, phase state, or mechanism for rotating one coordinate into the other.

3 The installed Mamba-3 SISO rotational path

This section describes the code shipped in the installed `mamba-ssm 2.3.2.post1` package, not an inferred variant. Its input projection is ordered

$$[z, x, B, C, dd_{dt}, dd_A, \mathrm{trap}, \mathrm{angle}]. \tag{4}$$

The dd_{dt}, dd_A, and trapezoid fields each have H rows. In contrast, the angle field has only P_M rows, where P_M is half of the selected half- or full-rotary state prefix. The raw angle rows are expanded across heads. Thus Mamba-3 separates a token/head clock from token/pair angular content:

$$DT_{th} = \operatorname{softplus}(dd_{dt,th} + b^d_h), \tag{5}$$

$$A_{th} = \min\{-\operatorname{softplus}(dd_{A,th}), -A_{\mathrm{floor}}\}, \qquad \rho_{th} = \exp(A_{th} DT_{th}), \tag{6}$$

$$\nu_{tj} = \pi \tanh(e_{tj}), \tag{7}$$

$$\Delta\theta_{thj} = DT_{th} \nu_{tj}, \tag{8}$$

$$\theta_{thj} = \theta_{0,hj} + \sum_{s=1}^{t} \Delta\theta_{shj} pmod 2\pi. \tag{9}$$

The cumulative phase is positive. For every selected adjacent pair, the SISO kernel applies the same $R(+\theta_{thj})$ to Q_t and K_t. Consequently the same-token dot product is unchanged, whereas a cross-token term obtains the relative rotation

$$(R(\theta_t) q_t)^{\mathsf{T}} R(\theta_s) k_s = q_t^{\mathsf{T}} R(\theta_s - \theta_t) k_s. \tag{10}$$

If S_t denotes a state in a fixed physical frame, this is equivalent to the moving-frame recurrence

$$S_t^- = \rho_t R(-\Delta\theta_t) S_{t-1}, \tag{11}$$

pair by pair, with scalar radial decay ρ_t. The identity follows from $R(\theta_t)^\top R(\theta_{t-1}) = R(-\Delta\theta_t)$. It is the key computational idea: no dense rotational state transition is needed when radial coefficients are pair-isotropic.

Mamba-3 additionally uses a trapezoidal SSM write. With

$$\gamma_t = DT_t \ \mathrm{sigmoid}(\mathrm{trap}_t), \qquad \eta_{t+1} = DT_{t+1}\big(1 - \mathrm{sigmoid}(\mathrm{trap}_{t+1})\big), \tag{12}$$

the offline kernel scales token t's rotated key by $\gamma_t + \eta_{t+1}$, while its same-token contribution uses γ_t. At a streaming boundary, the cached previous key/value contributes the missing η_t term. This is a two-endpoint discretization of the Mamba write; it is not the KLA innovation update defined below.

The angle clock is also coupled in backward. The VJP of the positive cumulative phase is a reverse cumulative sum. It produces both

$$\bar{e}_{tj} = \sum_h \overline{\Delta\theta}_{thj}\, DT_{th}\, \pi\big(1 - \tanh^2(e_{tj})\big), \tag{13}$$

$$\overline{DT}^{\mathrm{angle}}_{th} = \sum_j \overline{\Delta\theta}_{thj}\, \pi \tanh(e_{tj}). \tag{14}$$

Here the sum over h is the backward of the head expansion of the shared raw angle rows. The combined implementation explicitly adds $\overline{DT}^{\mathrm{angle}}$ to the DT gradient from the radial SSM dynamics. This is a coherent choice for a shared physical discretization clock, but it is intentionally not the RDiag E2 gradient contract.

4 RDiag KLA with the E2 bounded clock

4.1 Pair-tied radial dynamics

Let the radial preactivation be f_{bthk} with bias b^d_{hk}, and let

$$d_{bthk} = \mathrm{softplus}(f_{bthk} + b^d_{hk}). \tag{15}$$

For rotary pair j, define the pair step and tied process noise

$$\bar{d}_{bthj} = \tfrac{1}{2}(d_{bth,2j} + d_{bth,2j+1}), \tag{16}$$

$$\bar{\omega}_{bthj} = \tfrac{1}{2}(\omega^{\mathrm{in}}_{bth,2j} + \omega^{\mathrm{in}}_{bth,2j+1}), \tag{17}$$

$$\alpha_{bth,2j} = \alpha_{bth,2j+1} = \exp\Big[-\exp(A^{\log}_h)\bar{d}_{bthj}\Big], \tag{18}$$

$$\omega_{bth,2j} = \omega_{bth,2j+1} = \bar{\omega}_{bthj}. \tag{19}$$

Only the selected prefix is pair-tied; tail lanes retain the diagonal-KLA frontend's original α and ω. Pair tying is not cosmetic. On each pair it makes $\mathrm{diag}(\alpha_t) = \alpha_{tj} I_2$ and the diagonal process covariance $\mathrm{diag}(\omega_t) = \omega_{tj} I_2$, so the *prediction* terms commute with $R(\vartheta)$. It does not force posterior information or the Kalman gain to be pair-isotropic; an observation generally splits them. The diagonal posterior approximation is defined in the current moving frame, as made explicit below.

4.2 Detached live radial clock and persistent anchor

Set the open upper clock bound to $C = 0.1$. At construction, snapshot one FP32 anchor per head,

$$a_h = \mathrm{mean}_k\, \mathrm{softplus}(b^d_{hk}), \qquad 0 < a_h < C. \tag{20}$$

The anchor is a persistent buffer, not a parameter. At each token compute the detached live radial mean

$$m_{bth} = \mathrm{mean}_k \,\mathrm{softplus}\Big(\mathrm{stopgrad}(f_{bthk} + b^d_{hk})\Big)\,, \tag{21}$$

and a learnable token/head residual

$$z^{\mathrm{clk}}_{bth} = \frac{\langle w_h, x_{bt}\rangle}{\sqrt{D}} + \beta_h, \tag{22}$$

evaluated in FP32. The E2 clock is

$$\boxed{\tau_{bth} = C\ \mathrm{sigmoid}\Big[\log m_{bth} - \log(C - a_h) + \tanh(z^{\mathrm{clk}}_{bth})\Big]} \qquad (C = 0.1). \tag{23}$$

The bounded residual is strictly in $(-1, 1)$ and the sigmoid keeps a finite valid clock strictly in $(0, C)$. The implementation does not clamp or use a straight-through estimator: invalid, nonfinite, endpoint, or dead-derivative conditions are made visible to health checks, with invalid clock outputs replaced by NaNs.

Detachment is a model choice, not merely an optimization. The current radial timescale determines the forward phase scale, while the phase loss cannot change f, b^d, $A^{\log}$, α, or ω through this clock. The persistent anchor fixes the initialization reference across save/resume; recomputing it from a trained b^d would change the model after restart.

4.3 Angle, sign convention, and rotating basis

One projection shared across heads produces e_{btj}. The head-specific increment is

$$\delta_{bthj} = \pi \tanh(e_{btj})\tau_{bth}, \qquad |\delta_{bthj}| < 0.1\pi. \tag{24}$$

RDiag uses the negative inclusive phase scan

$$\phi_{bthj} = \phi_{b0,hj} - \sum_{s=1}^{t} \delta_{bshj}, \tag{25}$$

implemented in FP32 and reduced to $[0, 2\pi)$ at the bounded-clock consumer. The same $R(+\phi_{thj})$ is applied to the corresponding pairs of q_t and k_t. Hence, for $s < t$,

$$(R(\phi_t)q_t)^{\mathsf{T}} R(\phi_s)k_s = q_t^{\mathsf{T}} R\left(\sum_{u=s+1}^{t} \delta_u\right) k_s. \tag{26}$$

Under the column-vector convention of Eq. (1), the negative scan therefore represents $R(+\delta_t)$ transport. Mamba-3's positive scan represents $R(-\Delta\theta_t)$. This sign is a gauge choice because replacing the zero-initialized angle projection by its negative maps one convention to the other. It is nevertheless an implementation invariant: forward, streaming carry, and the negative suffix-scan VJP must use the same sign.

More explicitly, let $\mathcal{R}_t$ be block diagonal with the pair rotations $R(\phi_{thj})$ and an identity tail. The corresponding physical transition and process covariance are

$$D_t = \mathcal{R}_t^{\mathsf{T}} \,\mathrm{diag}(\alpha_t)\mathcal{R}_{t-1}, \qquad \Omega^{\mathrm{phys}}_t = \mathcal{R}_t^{\mathsf{T}} \,\mathrm{diag}(\omega_t)\mathcal{R}_t. \tag{27}$$

On a tied pair, these reduce to $D_{tj} = \alpha_{tj} R(+\delta_{tj})$ and $\Omega^{\mathrm{phys}}_{tj} = \omega_{tj} I_2$. With moving-frame state $\bar{S}_t = \mathcal{R}_t S_t$ and addresses $\bar{k}_t = \mathcal{R}_t k_t$, $\bar{q}_t = \mathcal{R}_t q_t$, the mean-state recurrence becomes purely radial and can reuse the diagonal-KLA memory kernel.

4.4 Diagonal Kalman gain and innovation memory

Let the phase-rotated addresses be $\widetilde{q}_t = \bar{q}_t$ and $\widetilde{k}_t = \bar{k}_t$. With posterior diagonal information $c_{t-1,hk} > 0$, observation noise $r_{th} > 0$, and information scale $s_h > 0$, RDiag KLA uses the moving-frame state (written as S_t below for brevity)

$$p_{thk}^{-} = \frac{\alpha_{thk}^2}{c_{t-1,hk}} + \omega_{thk}, \tag{28}$$

$$\kappa_{thk} = \frac{p_{thk}^{-}\widetilde{k}_{thk}}{r_{th} + \sum_i p_{thi}^{-}\widetilde{k}_{thi}^2}, \tag{29}$$

$$c_{thk} = \frac{1}{p_{thk}^{-}} + s_h \frac{\widetilde{k}_{thk}^2}{r_{th}}. \tag{30}$$

For associative state $S_{th} \in \mathbb{R}^{K \times V}$,

$$S_{th}^{-} = \operatorname{diag}(\alpha_{th}) S_{t-1,h}, \tag{31}$$

$$S_{th} = S_{th}^{-} + \boldsymbol{\kappa}_{th} \left(\boldsymbol{v}_{th} - \widetilde{\boldsymbol{k}}_{th}^{\mathsf{T}} S_{th}^{-} \right)^{\mathsf{T}}, \tag{32}$$

$$o_{th} = S_{th}^{\mathsf{T}} \left(K^{-1/2} \widetilde{\boldsymbol{q}}_{th} \right). \tag{33}$$

This is the KLA estimator. It is not a Mamba SSM write: κ_t is an adaptive anisotropic Kalman gain and the write is the current innovation. The diagonal information recurrence can be computed by its existing associative Möbius scan, followed by the existing chunked affine memory recurrence. In physical coordinates, c_t represents the model's diagonal projection of $\mathcal{R}_t P_t^{-1} \mathcal{R}_t^{\mathsf{T}}$; it is not a claim that rotating an arbitrary anisotropic posterior covariance remains diagonal. In particular, c, κ, k, and q are never pair-tied.

4.5 VJP isolation

The computational graph has two intentionally separate branches:

$$(f, b^d, A^{\log}, \omega^{\text{in}}) \longrightarrow (\alpha, \omega) \longrightarrow \text{Kalman dynamics},$$
$$\operatorname{stopgrad}(f + b^d),\ a,\ (W, \beta),\ e \longrightarrow \tau, \delta, \phi \longrightarrow \text{rotated } q, k.$$

In its radial-only specialization, the direct-pair backward has no angle data dependency or load. Conversely, the clock's m and a arguments are detached, so the phase VJP returns no gradient to the radial gate, radial bias, or $A^{\log}$. Unlike Mamba-3, RDiag does not add an angular-clock contribution to the dynamical radial-step gradient. Among the new phase-specific parameters, gradients reach the angle projection and clock weight/bias; they also return to the shared hidden states through both projections. The isolation claim is specifically that no clock edge reaches the radial parameters.

5 What to take from Mamba-3

The useful transfer from Mamba-3 is structural: encode rotation as a relative query/key phase with a small shared angle projection and a token/head clock. The estimator-specific pieces should remain KLA-specific.

Aspect	Mamba-2 SISO	Mamba-3 SISO	RDiag KLA E2
Transition geometry	Scalar radial decay; no rotation	Radial SSM plus relative pair rotation	Pair-tied radial KLA plus relative pair rotation
Clock	$DT_{th} > 0$	$DT_{th} > 0$ shared by decay and angle	$0 < \tau_{th} < 0.1$, based on detached live radial mean plus residual
Angle rows	None	Token/pair rows shared across heads	Token/pair rows shared across heads; head-specific τ
Phase gauge	None	Positive cumulative phase	Negative cumulative phase; sign documented by Eq. (26)
Write	Euler-like $DT\,B\,x$	Trapezoidal two-endpoint SSM write	Kalman innovation with anisotropic gain
Clock gradient	Only SSM dynamics	Angle dDT is added to dynamical dDT	Phase/radial VJPs intentionally isolated
Covariance	No Kalman posterior covariance	No KLA diagonal information update	Diagonal information in moving frame; pair-isotropic prediction parameters

Concretely, RDiag should take the following from Mamba-3:

1. represent rotation by the same adjacent-pair transform on queries and keys, avoiding a dense recurrent rotation;
2. use a bounded angular rate $\pi \tanh(e)$ and one token/head clock broadcast over pair modes;
3. share the compact angle projection across heads, while permitting head-specific timing;
4. accumulate phase and streaming carry in FP32, with a specified modular reduction contract;
5. treat every dependency of a shared clock explicitly in backward and test cancellation-heavy reductions against the executable reference; and
6. retain half- or full-prefix rotary layouts rather than forcing every key lane to rotate.

It should intentionally *not* copy the following:

1. Mamba-2's scalar-only transition as a purported rotational mechanism;
2. Mamba-3's trapezoidal SSM write, because KLA's innovation update defines a different estimator;
3. Mamba-3's angle-to-DT gradient coupling, because E2 deliberately uses a detached radial timescale;
4. approximate softplus, trigonometric, or reassociated reduction sequences without proving the required finite-precision parity;
5. independent radial coefficients inside a rotated pair while claiming a literal scalar-damped rotation; they instead define a more general rotated anisotropic contraction; or
6. hard clock clamps or straight-through boundary gradients, which conceal saturation instead of making it observable.

6 Implementation contract

6.1 Layout and forward pipeline

The production-oriented route is restricted to contiguous CUDA tensors, BF16 projected $q, k, v, f_{\mathrm{raw}}, z_{\mathrm{raw}}, y_{\mathrm{raw}}$, FP32 frontend parameters, equal key/value head counts, and $K \in \{64, 128\}$.

The clock preactivation z^{clk} is a separate FP32 quantity. The rotary fraction is 1/2 or 1, so K_r is positive and even. Adjacent memory lanes $(0,1),(2,3),\ldots,(K_r-2,K_r-1)$ form the rotary pairs; no split-half pairing is permitted.

The forward pipeline is:

1. project q,k,v and the diagonal-KLA radial/noise quantities;
2. compute the standard frontend's frozen α^{in} and ω^{in};
3. call the unchanged direct-pair operator with a detached angle, retain its pair-tied α, ω, and log-decay prefix, and discard its legacy phase output;
4. independently materialize the precise detached FP32 $[B,T,H,K]$ clock steps and form the exact native-framework radial mean m described below;
5. keep Wx as the existing FP32 GEMM, then fuse the pointwise clock chain and Eq. (24), using the exact CUDA-eager sigmoid quotient;
6. perform the bounded negative phase scan in chunks of 64, retaining an FP32 carry;
7. normalize q,k in FP32, round the memory-facing copies as specified, and rotate adjacent pairs by the same phase; and
8. run the diagonal information/gain kernel and the S3 chunked KLA memory recurrence.

The gain-facing key remains FP32. The memory-facing query/key route is rounded through BF16 in the same place as its executable reference before rotation/final rounding; changing that boundary is a numerical model change for parity purposes.

6.2 The precise detached-clock mean

The actual radial dynamics retain the FLA softplus used by the qualified baseline. The detached clock uses a separate precise FP32 copy

$$\text{softplus}_{20}(x) = \begin{cases} x, & x > 20, \\ \log 1p(\exp x), & x \leq 20, \end{cases} \tag{34}$$

evaluated with the precise device intrinsics. To reproduce the eager reference, the optimized path must materialize

$$\text{clock\_dt} \in \mathbb{R}^{B\times T\times H\times K} \tag{35}$$

in FP32 and then invoke the framework reduction

$$\text{radial\_clock} = \texttt{torch.mean(clock\_dt, dim=-1, out=radial\_clock)}. \tag{36}$$

A mathematically equivalent Triton lane sum is not an exact substitute. It changes addition order and was shown to leave small residual differences that can be amplified by long phase scans and cancellation-heavy angle gradients. The precise clock copy is detached; the FLA-softplus radial outputs and their VJP remain untouched. The temporary costs exactly $4BTHK$ bytes. At $B=4, T=4096, H=16$, that is 64 MiB for $K=64$ and 128 MiB for $K=128$; at $H=18, K=128$ it is 144 MiB. This cost belongs in the end-to-end memory and speed gate rather than being hidden by a numerically different reduction.

6.3 Reusing the direct-pair radial path safely

The production wrapper does not maintain a second implementation of Eq. (19). It calls the established direct-pair transform with `angle_raw.detach()`, ignores the returned legacy delta, and reuses only g, α, and ω. This keeps the radial forward arithmetic and its VJP identical to the already-qualified direct-pair path.

Discarding the legacy delta makes its incoming cotangent absent. The direct-pair backward turns that fact into the compile-time specialization `HAS_DDELTA=false`. In this specialization the generated kernel does not load the detached angle, evaluate its tanh, or form the phase term $\pi \tanh(e)\,\overline{\delta}$. The guard is semantically important: merely setting $\overline{\delta} = 0$ would not isolate a nonfinite discarded angle, because IEEE arithmetic gives $0 \times \mathrm{NaN} = \mathrm{NaN}$. The phase-enabled `HAS_DDELTA=true` algebra is unchanged, while dedicated NaN/$\pm\infty$ tests establish that a discarded phase cannot contaminate radial outputs or gradients.

6.4 Exact sigmoid and bounded-delta forward

Let

$$\ell_{bth} = \log m_{bth} - \log(C - a_h) + \tanh(z^{\mathrm{clk}}_{bth}). \tag{37}$$

For FP32 CUDA inputs, PyTorch's native sigmoid kernel evaluates $1/(1 + \exp(-\ell))$. The generic Triton `tl.sigmoid` uses a different approximate lowering and produced one-ULP clock differences even when ℓ was bitwise identical. The production kernel therefore spells out the eager expression as

$$\tau^{\mathrm{raw}}_{bth} = 0.1\,\texttt{tl.div\_rn}\left(1,\, 1 + \texttt{libdevice.exp}(-\ell_{bth})\right), \tag{38}$$

then applies the fail-visible validity predicate to obtain τ. The angle nonlinearity is computed once by native ATen as `torch.tanh(angle_raw.float()).contiguous()` and passed to the fused clock kernel, which forms $\delta = \pi \tanh(e)\tau$. This boundary makes the actual-logit τ and δ bitwise equal to the eager oracle, rather than merely close in real arithmetic.

6.5 Phase scan and backward order

The phase forward is an inclusive FP32 scan. Each chunk computes $\phi = \mathrm{carry} - \mathrm{cumsum}(\delta)$, wraps valid values to $[0, 2\pi)$, and passes the wrapped final value as the next carry. Before reduction, $|\phi|$ must not exceed 65,536; nonfinite, out-of-contract, or noncanonical residues become sticky NaNs. Its VJP is the global negative suffix scan

$$\overline{\delta}_t = -\sum_{u=t}^{T} \overline{\phi}_u, \tag{39}$$

with the final-carry cotangent subtracted from every token when present.

For the fused clock/delta backward, operation order is part of the contract. Let $g_{thj} = \overline{\delta}_{thj}$. The eager-compatible order is

$$\overline{\tau}_{th} = \sum_j \left[g_{thj} \big(\pi \tanh(e_{tj}) \big) \right], \tag{40}$$

$$\overline{z}^{\mathrm{clk}}_{th} = \mathrm{mask}_{th}(\overline{\tau}_{th})\, \tau^{\mathrm{raw}}_{th} \left(1 - \frac{\tau^{\mathrm{raw}}_{th}}{C}\right) \left(1 - \tanh^2 z^{\mathrm{clk}}_{th}\right), \tag{41}$$

$$\overline{e}_{tj} = \left[\left(\sum_h g_{thj} \tau_{th} \right) \pi \right] \left(1 - \tanh^2 e_{tj}\right). \tag{42}$$

Here $\text{mask}_{th}(g)$ is g when the forward validity predicate selected τ_{th}^{raw} and zero when it selected the detached NaN branch. The subsequent multiplication order is retained, so a NaN local derivative remains fail-visible even after a zero cotangent mask. Equation (40) must be evaluated as the displayed broadcast multiply followed by the framework's sum over P; pulling π outside the sum is not accepted as "equivalent." For Eq. (42), multiply by τ, reduce over H, multiply by π, and only then apply the tanh derivative. Concretely, production saves the native-ATen tanh output from the forward and evaluates

```
torch.ops.aten.tanh_backward( (d_delta * tau[..., None]).sum(dim=2) * pi, angle).
```

The result is cast to `angle_raw`'s dtype (BF16 on the production route) before the angle projection VJP. No Triton angle-partial tensor or head-reduction kernel remains. The validity mask is applied to the incoming clock cotangent before the sigmoid/tanh clock chain, matching the eager `where(valid, raw_tau, detached_nan)` VJP. Weight and hidden-state gradients then flow through the ordinary linear operations.

6.6 Persistent state and FSDP

The anchor a_h must satisfy all of the following:

1. FP32, contiguous, shape $[H]$, finite, and strictly inside $(0, 0.1)$;
2. registered as a persistent buffer and present in the model state dictionary;
3. absent from named parameters and optimizer groups;
4. identical across distributed ranks after construction, FSDP wrapping, save, and resume;
5. restored from the checkpoint rather than regenerated from the current radial bias.

Production enforces this with `bounded_rdiag_fsdp_mixed_precision`: it retains the installed BF16-mixed parameter/reduction policy but overrides `buffer_dtype=torch.float32`. Resume tests compare the anchor bytes before and after round trips and verify that fresh construction cannot silently overwrite a checkpointed anchor.

6.7 Initialization and health signals

The angle projection weight and bias, and the bounded-clock residual weight and bias, are initialized to zero. Rotation therefore starts at the *pair-tied* diagonal-KLA function: $e = 0$ gives $\delta = 0$ even when τ is nonzero, while the pair averaging in Eq. (19) remains active. This is preferable to initializing the clock at zero, which would place it on a boundary and kill useful clock gradients once angles become live.

The audited bounded RDiag named configurations use $r_{\min} = 0.01$, process-noise floor $\omega_{\min} = q_{\min} = 0$, and `gain_init=iso`. The last setting zeroes the output map of the process-noise projection, so ω starts uniform across key channels; it does *not* initialize ω itself to zero, because the positive softplus remains. These are recipe and initialization choices rather than changes to Eqs. (28)–(33).

At every checked optimizer boundary, diagnostics should aggregate across microbatches, layers, and ranks:

- nonfinite counts for the radial mean, base logit, residual, logit, raw clock, valid clock, derivative, phase, parameters, and gradients;
- exact endpoint and near-boundary counts, using 10^{-4} and $0.1 - 10^{-4}$ as the observed lower and upper near-boundary bands;

- minimum positive clock Jacobian and a near-dead count below 10^{-8};
- token variation of the residual projection after bootstrap;
- phase magnitude, delta magnitude, and canonical wrapped-range checks; and
- schema/count agreement across ranks, including exactly one observation per expected micro-batch/layer pair.

The kernel itself fails visibly at invalid endpoints. Promotion screens may impose stricter distributional thresholds than the continuous health controller; both classes of gate are needed.

7 Reference pseudocode

The following pseudocode fixes the mathematical order without prescribing launch geometry.

1. **Radial and clock preparation.**

$$\begin{aligned} x^d &= f + \texttt{dt\_bias}, \\ d^{\text{radial}} &= \texttt{FLA\_softplus}(x^d), \\ \texttt{clock\_dt} &= \text{softplus}_{20}(\text{stopgrad}(x^d)) \quad \text{materialized FP32 } [B, T, H, K], \\ m &= \texttt{torch.mean(clock\_dt, dim=-1, out=radial\_clock)}. \end{aligned}$$

   Invoke the unchanged direct-pair transform with a detached angle, pair-average d^{radial} and ω on the rotary prefix, and construct α from Eq. (19). Discard its legacy delta and do not use `clock_dt` for radial dynamics.

2. **Bounded clock and phase.**

$$\begin{aligned} z^{\text{clk}} &= \texttt{linear}(x, W)/\sqrt{D} + \beta, \\ \ell &= \log m - \log(0.1 - a) + \tanh z^{\text{clk}}, \\ \tau &= 0.1\, \texttt{div\_rn}(1, 1 + \texttt{libdevice.exp}(-\ell)), \\ \widehat{e} &= \texttt{torch.tanh}(e\texttt{.float()}), \\ \delta &= \pi \widehat{e}\, \tau, \\ \phi_t &= \text{wrap}_{2\pi}(\phi_{t-1} - \delta_t). \end{aligned}$$

   Reject invalid open-clock or phase-contract values by producing observable NaNs.

3. **Addresses and Kalman recurrence.** Rotate the same adjacent prefix of normalized q and k by $R(\phi)$, preserve tail lanes, then apply Eqs. (28)–(33). The chunked implementation must match this token recurrence under the documented BF16/FP32 boundary.

4. **Backward.** Run Eq. (39); use Eqs. (40)–(42) exactly; evaluate the angle derivative with native `aten.tanh_backward` on the saved $\widehat{e}$; compile the discarded direct-pair phase branch with `HAS_DDELTA=false`; return no clock gradient to m or a; and run the independent radial/Kalman VJPs.

8 Lessons from the E-screen and performance/numerical debugging

An open interval is not a distributional guarantee. An earlier bounded-clock E screen completed 519 optimizer updates and produced no exact endpoints or nonfinite clocks, yet the step-520 evaluator rejected it. In layer 0, the lower-near-boundary fraction was 0.00205326, above the preregistered 10^{-3} threshold; the observed clock range was approximately $[8.65 \times 10^{-5}, 0.09813]$. Thus a smooth sigmoid bound alone did not prevent probability mass from accumulating near a boundary. The design response was the detached live radial base plus a bounded token/head residual, not a harder clamp.

The redesigned E2 screen reached its step-520 evaluation after 519 completed optimizer updates (520 measured gradients), with zero lower and upper near-boundary counts in the reported gate and an observed clock range of approximately [0.01060, 0.08181]. That result supports the clock design and justified integration work. It does not prove long-run training quality, kernel parity under every shape, distributed resume correctness, or a production speedup.

Correctness-first composition exposed the actual performance target. At $B = 4, T = 4096, H = 16, K = V = 128, D = 2048$, the unfused rotational-v1 wrapper measured 24.706 ms for combined mixer forward/backward versus 13.735 ms for its frozen control, a 1.799× ratio (79.9% overhead). The identical-input frozen S3 stage was 0.999× its control, localizing the cost to the eager second-dt, pair tying, phase/trigonometry, three rotations, and prefix rebuild. This motivated the direct radial kernel, fused normalization/rotation, and fused clock/delta path while retaining the established gain and S3 kernels. These are historical isolation numbers for the v1 geometry, not a speed claim for the final E2 composition. The later final A/B measurements, including the exact-mean FP32 scratch, are reported in Table 1.

A tiny local error can become a large gradient discrepancy. The first optimized clock path reused the FLA radial softplus. Against eager PyTorch, the initial radial-mean discrepancy was only about 6.33×10^{-8} in maximum absolute value and 1.54×10^{-6} in relative L_2. After the logit, long phase scan, BF16 rotation boundary, and cancellation across heads, the angle-weight gradient showed about 3.81×10^{-3} relative L_2 error in the focused diagnostic. Near-zero reference entries also made elementwise relative metrics misleading, so absolute, normwise, and cancellation-stress gates are all required.

Using Eq. (34) reduced the clock error substantially, but a Triton reduction still did not reproduce eager bitwise. Materializing the FP32 $[B, T, H, K]$ precise values and using the exact framework mean was bitwise identical in the focused $K = 64$ and $K = 128$ experiment while leaving the FLA radial α, log-prefix, and ω outputs bitwise unchanged. Likewise, algebraically reassociated all-Triton backward reductions failed the focused VJP gate; the explicit framework reduction of Eq. (40) and the factor order of Eq. (42) passed it. The lesson is narrow but important: the executable eager graph, including reduction order, is the reference for this mixed-precision path.

Elementary functions are also part of the executable reference. After the exact mean fix, a full $K = 64$ trace found bitwise-identical radial clock, base logit, bounded residual, and final logit, yet the fused clock still differed from eager at 61,655 of 262,144 FP32 τ entries (maximum absolute error 7.45×10^{-9}). Those one-ULP clock differences propagated to 929,205 of 4,194,304 delta entries and were amplified by the length-4096 phase scan. The cause was `tl.sigmoid`, not the model equations. Replacing it with Eq. (38), matching PyTorch's CUDA source expression and rounded division, removed the actual-logit discrepancy. Computing the angle tanh and its backward with native ATen then removed the remaining reduction/transcendental-order drift. This is why the final path uses selective eager operations at two small boundaries rather than insisting that every operation be fused into Triton.

9 Invariants and qualification status

The following are invariants, not tunable approximations:

1. rotary prefix width is positive and even; pairing is adjacent;
2. α and ω are exactly equal within each selected pair;
3. q and k receive the same phase and same rotation matrix;
4. the phase scan is negative and inclusive, and its backward is a negative suffix scan;
5. m and a are detached FP32 quantities; phase gradients do not enter radial dynamics;
6. the valid clock and its local derivative are finite and strictly positive, with $0 < \tau < 0.1$;
7. the anchor is persistent FP32 checkpoint state, not optimizer state;
8. precise clock softplus plus `torch.mean` is separate from FLA radial softplus;
9. the bounded sigmoid uses Eq. (38), not `tl.sigmoid`;
10. the displayed forward and backward reduction orders are preserved, with the angle tanh and its VJP evaluated by native ATen;
11. a discarded direct-pair delta selects `HAS_DDELTA=false`, eliminating every angle access and phase product from the radial backward; and
12. the unrotated tail is an exact pass-through of diagonal KLA.

The one-H200 kernel/layer checklist is closed by the cumulative qualification campaign:

1. pure-PyTorch token recurrence versus optimized forward for zero, half, and full rotation;
2. first-order VJPs for every input/parameter, including adversarial cancellation and partial chunks, at $K = 64$ and $K = 128$;
3. exact checks for radial outputs and detached clock mean where exactness is part of the contract, plus justified tolerances elsewhere;
4. segmented/streaming equivalence, phase carry, wrap-boundary, and sticky-NaN tests;
5. zero-angle equivalence to pair-tied diagonal KLA and angle-sign/gauge unit tests;
6. multi-step health checks covering bootstrap-zero angle, subsequently live clock gradients, token variation, and no nonfinite/endpoints;
7. exact full-layer forward/VJP comparison with the sealed E2 candidate at $K = 64$, $K = 128$, and the 18-head 1.3B-style geometry; and
8. interleaved A/B timing against both the sealed E2 candidate and the legacy pair clock.

The authoritative one-H200 job 1777777 completed with exit code 0 and reported 117 passing core RDiag tests followed by 54 passing E2 tests. Table 1 gives the paired runtime ratios for an exact RDiag-layer forward plus backward; values below one are faster than the named reference. Every upper endpoint of the bootstrap 95% confidence interval is below the 1.05 promotion limit.

Thus the optimized E2 implementation is bitwise faithful where exact parity is required, faster than the sealed accepted implementation in all three measured geometries, and within roughly 1–1.4% of the simpler legacy clock. This closes the E2 one-H200 kernel/layer gate, but not a full production-submission gate. Train-smoke, distributed checkpoint/resume, and multi-node

Table 1: Final H200 paired A/B timing from job 1777777, with 40 measured rounds after 20 warmups. "Accepted" is the sealed eager E2 candidate and "legacy" is the production pair-dt path.

Geometry (H, D, K)	E2/accepted	95% CI high	E2/legacy	95% CI high
$(16, 1024, 64)$	0.969512	0.978247	1.014135	1.020172
$(16, 2048, 128)$	0.950407	0.956546	1.008590	1.013666
$(18, 2304, 128)$	0.960111	0.964929	1.013622	1.014984

soak/attestation remained for that baseline at the time of sealing. The next section records the later shared-DT candidate and its separate, still-open qualification.

10 Shared-DT engineering status as of September 4, 2026

This section records a newer production candidate and does not revise the closed E2 result above. The candidate starts from commit `6472a0711eb61177c3ef91845855fffa82202290` and replaces E2's detached bounded clock with a shared-DT transition. Its qualification is *not complete*. Four gates passed in the first campaign, whose gradient gate exposed both a defective acceptance rule and real rotational-gradient amplification; that soak was cancelled before allocation. The subsequent reviewed head-scaled commit passed all five non-soak gates, but its 3300-update soak failed closed with an aggregate count at boundary 269. A divergent diagnostic replay independently exposed rejection of the finite-input exact-zero decay endpoint. The narrow lower-endpoint correction then passed all five non-soak gates and 675 soak updates before failing closed at boundary 676 with an unrecorded key. A second divergent replay independently exposed rejection of the finite-input exact-one endpoint; the reviewed upper-endpoint correction passed its fresh numerical H200 suite, but its evidence could not be sealed because a pre-existing asynchronous log-finalization race omitted the terminal PASS marker from the retained log. The narrow source-gate logging correction was reviewed, committed, and published. Its fresh campaign passed and sealed the H200 gate and all four short distributed gates, then its soak crossed both old boundaries and failed closed at update 790 on a rank-equal pre-clip global norm that triggered the combined nonfinite, negative, or greater-than-10 rejection predicate. The failed run did not retain which branch fired. No soak seal or aggregate authorization exists; the diagnostic replay and threshold-provenance analysis are recorded below.

10.1 Relation to Mamba-2 and Mamba-3

The source-level comparison in Sections 2 and 3 found no discrepant transition equation or kernel sign bug in the shared-DT path. Its radial factor is the Mamba-2 form

$$\alpha_{th} = \exp[-\exp(A_{\log,h})DT_{th}]\,, \tag{43}$$

and its bias-free pair-angle projection follows the installed Mamba-3 construction,

$$\delta_{thj} = \pi \tanh(e_{tj})DT_{th}, \qquad \phi_{thj} = -\sum_{s\le t} \delta_{shj}, \tag{44}$$

with one angle value per token/pair broadcast over heads, a headwise DT, a cumulative phase scan, and the corresponding negative reverse-suffix backward. This is the literal RDiag source convention; installed Mamba-3 uses the positive scan. The difference is the exact gauge $W_{\text{angle}} \mapsto -W_{\text{angle}}$ and therefore cannot explain a gradient-norm change. The relevant difference from the E2 model is instead that the angle and DT paths remain live and the same shared DT controls radial decay and rotation.

10.2 Issue, diagnosis, and completed work

Two distinct problems were found. First, the original per-microbatch global-gradient limit of 10 was not a valid gate: all 24 measured model/batch observations exceeded 10, including stock DiagKLA at 14.001–14.232. Stock DiagKLA and the bounded-clock layer were also architecture-mismatched controls. Second, the shared angle projection creates genuine coherent amplification when its output is broadcast over all recurrent heads and differentiated through the cumulative rotation. This is a model-parameterization issue, not evidence of a faulty radial or phase kernel.

On the same eight real FineWeb batches, the unmodified shared-DT model produced global norms 34.596–45.300 (median 38.758), versus a DiagKLA median of 14.096. Disabling rotation returned the median to 14.180, while detaching only the angle path gave 18.035 and zero angle initialization still gave 25.593. A phase-only DT cap of 0.1 was insufficient. Scaling the effective angle preactivation by $1/\sqrt{16} = 0.25$ gave median 16.765 and shared/Diag median and 95th-percentile nearest-rank ratios 1.191 and 1.261, with live angle and DT gradients. These interventions isolate head-shared cumulative rotation as the amplification source.

Completed engineering work. The Mamba-2/Mamba-3 source audit, exact H200 measurement, rotation/gradient ablations, matched- backbone control, and scale/cap sweep are complete. The replacement design and implementation plan received an independent review with no remaining Important findings. The layer-side $H^{-1/2}$ change and its nonpersistent-state host contracts are now implemented in the isolated checkout below: 41 host tests passed. Development H200 job 1827839 passed the focused H=18 all-parameter oracle/VJP case and H=16, T=4096 BF16 boundary/backward case; job 1827905 then passed the full 109-test production-H200 suite. These runs exercised the mutable development checkout, so they are correctness evidence rather than immutable qualification seals.

The same checkout now also contains the complete gradient-v3 control closure: both arms construct the same 220M configuration, strictly load one full reference state, prove matching configuration, parameter-inventory, and full-state digests, and reset CPU/CUDA randomness symmetrically. The no-rotation arm installs connected-zero hooks only on the twelve angle projections, checks all twelve angle-weight gradients are present, finite, and exactly zero for every batch, checks the shared radial gradient norms remain finite and positive, and proves hook removal. The v3 recipe, catalog digest, result attestor, stale-schema rejection, and aggregate-v3 recursive requirement are integrated. The focused gate-runner host suite passed 38 tests; the complete stable host sweep is also complete: the qualification-control suite passed 71 tests with one filesystem-capability skip (the skipped case passed separately on local temporary storage), and the broader thirteen-file shared-DT host sweep passed 418 tests with three expected skips.

In campaign `9870d362bcbb4ec3b9b3cafb13459ef8`, production-H200 (job 1824028), train-smoke (1824365), checkpoint-fault (1824367), and authenticated-resume (1824368) produced recursively valid seals with SHA-256 prefixes `ecc03259`, `9ca6354e`, `06c1f60a`, and `656c9412`, respectively. Gradient job 1824370 failed the invalid absolute-10 rule and produced no seal; the soak was cancelled before allocation. These seals characterize the unmodified base commit only and do not qualify the proposed head-scaled change.

Qualification outcome. The implementation commit passed separate exact-SHA specification/math and code-quality reviews without a finding and was pushed to the branch named below. Its immutable snapshot contains 4227 files and has source-manifest SHA-256 `5b6b80bf8c38f4c959ac1f908a10fc01a25ff6986e77f0ee1c0a35de6843fbd9`. Fresh campaign `a65918f1771446e7acfbffcc3e5c7dc3` is closed incomplete. Production-H200 gate job 1828250 passed all 109 tests and attestor job 1828252 published seal SHA-256 prefix `a05c99d0544a721a`. The other four short gates and their attestors passed. Gradient A/B job 1828811 measured shared/no-rotation median and 95th-percentile ratios 1.179 and 1.262, inside the 1.25/1.50 limits; attestor 1828820 published seal SHA-256 prefix `0eabab83a02ac397`. Checkpoint-fault jobs 1828813/1828823, authenticated-resume jobs 1828814/1828821, and train-smoke jobs 1828815/1828822 published seals with SHA-256 values prefixes `a3558a17796d427c`, `1ea7a7da40576d20`, and `c2e19737cff63305`, respectively. They are valid partial evidence but cannot authorize the commit without a valid soak seal.

The production soak did *not* pass. Job 1828812 trained normally through completed update 268 (loss 5.356, global gradient norm 1.101, and zero reported health failures), then failed closed at the pre-optimizer boundary for update 269 with aggregate health-failure count 158720. It therefore emitted neither a result nor a seal, and dependent attestor 1828824 was cancelled. The exception predated the detailed failed-counter breakdown. Its stage excluded observation/parameter/gradient coverage, gradient-nonfinite, and bootstrap-liveness failures, but could not distinguish the remaining forward-domain predicates from parameter non-finiteness or collective-structure disagreement. That ambiguity motivated the fail-closed diagnostic replays below.

The first diagnostic replay identified the candidate predicate. Non-authorizing job 1829026 ran on a different H200 node pair and observed its first exact FP32 zero earlier, at boundary 241, so the exact-boundary harness correctly exited nonzero rather than labeling the run a reproduction. Every one of the 16 ranks nevertheless reported the same sole failed key, `alpha_nonpositive_count=128`: one scalar $[B, T, H]$ decay site materialized over the $K = 128$ key width. All other failure counters were zero; the aggregate diagnostics included $\min \alpha = 0$, $DT_{\max} = 7.3498993$, and $\max |\tanh(e^{\text{eff}})| = 0.9377124$. The retained scheduler log is `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_sdt_soak269_diag_1829026.log`, with SHA-256 `5dc12cd3ea0ea52765495763b28bd0185b038c306bad6c333fabb537eb67e6ee`. This is evidence for the same candidate failure mode, not an exact reproduction of the original boundary-269 trajectory. The original aggregate count $158720 = 1240 \times 128$ is consistent with the same width replication, but its exception did not retain a per-key breakdown. Intermediate job 1829217, from commit `28531cf959304ab26b22789f2ea4ce082adbed13`, remained healthy through boundary 269 on yet another node pair and correctly failed closed without a summary when its search condition was not met. Its source-manifest and scheduler-log SHA-256 values are respectively `0dee86b8f758f14aabd78047e2d0d530475be2d29f99d27587c256233644c73f` and `142120433da0eddb643d6d6160974c756397ed730fe802c009fbd1eaed1b3924`.

Final non-authorizing search job 1829305 ran from diagnostic commit

`06eaf237f5be9f79ee8d00075b79dcc0e7a9e96a`.

It completed 277 optimizer updates and stopped before clipping or update 278 with status `SAME_FAILURE_MODE` and `authorizing=false`. All 16 ranks agreed on the sole production-predicate count `alpha_nonpositive_count=128`; removing the repeated key-width dimension leaves one unique scalar decay site. Every recorded $A_{\log}$, $\exp(A_{\log})$, and $\exp(A_{\log})DT$ value was finite. Their global ranges were respectively $[0.0472397, 2.7735384]$, $[1.0483732, 16.0152016]$, and $[0.000105659, 104.639008]$. The largest exponent exceeds the FP32 round-to-nearest zero threshold of approximately 103.972; 87.3365 is only the normal-to-subnormal onset. Thus finite inputs to $\exp[-\exp(A_{\log})DT]$ can produce the observed stored zero.

The retained summary is

`diagnostic_runs/soak269_replay_06eaf237f5be_1829305/`
`artifacts/replay-summary.json`,

with SHA-256 558a1dde9c842b95d6afaafb16c54994f8e75aae68b7fed0bfef11cfa4718fb5; its trajectory has SHA-256 eeb607f58f40a156e4fb42df899ef27f724a17c3ede5f85906395a4917692614. The scheduler log has SHA-256 c46c802ac2b73fa875a40699ce2ac96b2f32bff61c1e128a54625e8916264b03. The replay compared 268 original updates, retained nine additional updates, and did not exactly match the original prefix; it therefore establishes the floating-point endpoint and candidate failure mode, not an exact boundary replay. The harness temporarily exempted only `alpha_nonpositive_count` so that it could retain this evidence and still stopped before the update; its logged zero health-failure count is not a production pass.

The evidence classifies the reproduced zero-alpha endpoint as a health-policy false positive rather than a model or kernel math defect; because the original exception lacks a per-key breakdown, it does not prove that job 1828812 had identical causality. At original update 268, 583512 distinct scalar decay sites were already below the diagnostic 10^{-6} floor while loss and gradients remained finite. The S3 prefix clamps this range to 10^{-6} and assigns the below-floor branch zero cotangent; the Kappa recurrence is also defined at raw $\alpha = 0$. The narrow correction keeps `alpha_nonpositive_count` as telemetry, adds `alpha_negative_count` as the fatal domain predicate, and leaves the model's alpha and both recurrences unchanged. At commit `83118d2...`, alpha nonfinites and $\alpha \geq 1$ remained fatal.

One login-namespace preparation attempt failed closed on the intentionally strict live-data identity check, and the first compute-side preparation attempt (job 1828245) failed before manifest publication because its three mutable roots had not been pre-created. Preparation was rerun in the Slurm user namespace with those private roots fixed at mode 0700; neither failed attempt executed a numerical gate.

Diagnostic evidence. The primary metric artifacts are

- `diagnostic_runs/gradient_norm_6472a071_20260902T170943Z/metrics.json`, SHA-256 `9c6a8a96a297799a`;
- `diagnostic_runs/gradient_rotation_ablation_6472a071_20260902T183817Z/metrics.json`, SHA-256 `274abd1902e7c7d3`;
- `diagnostic_runs/gradient_matched_control_6472a071_20260902T185000Z/metrics.json`, SHA-256 `d908637092fdfc3e`; and
- `diagnostic_runs/gradient_angle_scale_6472a071_20260902T185100Z/metrics.json`, SHA-256 `e503ab386a9de79d`.

All are rooted at `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1`.

10.3 Approved implementation and remaining qualification

Chosen change. For H recurrent heads, the development layer derives the nonpersistent scalar $H^{-1/2}$ and forms

$$e_{tj}^{\mathrm{eff}} = H^{-1/2}(W_{\mathrm{angle}} x_t)_j, \qquad \delta_{thj} = \pi \tanh(e_{tj}^{\mathrm{eff}}) DT_{th}. \tag{45}$$

The projection remains bias-free and its stored weight is unchanged. In the BF16 production path, the layer performs the multiply on the BF16 projection output before the frontend's FP32 cast and FP32 `tanh`; the frontend must not scale a second time. The scale is neither a parameter nor a buffer, so old state dictionaries remain strictly loadable. This preserves the transition family and both learning paths while normalizing the initial head-summed VJP.

The replacement gradient gate is an exact same-model/same-state two-arm intervention. The `shared_dt` arm runs normally; `shared_dt_no_rotation` replaces only each angle- projection output by the connected zero `output * 0.0`. Config, full state, parameter inventory, and radial shared-DT path must be identical. Acceptance requires shared/no-rotation nearest-rank median ≤ 1.25 and 95th percentile ≤ 1.50, both inclusive. The microbatch runaway sentinel becomes an inclusive 100; the late-training soak limit of 10 is unchanged. Only the gradient sample/result/recipe/seal closure advances to schema v3. Aggregate-v3 retains its schema name but must recursively require the v3 gradient seal.

Head-scale checkout and plans. Implementation is isolated from the main checkout in this worktree:

> `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_worktrees/rdiag_shared_dt_headscale_20260902`.

The observed checkout is branch `codex/rdiag-shared-dt-angle-headscale-20260902`, based on the commit above. The clean reviewed and published implementation snapshot is

> `a76dc7b930ae4d85157391d444c40ef7db14ef86`,

with tree `4cdb6287080365089349f56b14f1d782202d6157`. The reviewed design/plan snapshot is commit

> `9e621b5ede53529df50619804f0c9d8d32707284`.

Its documents are

- `docs/plans/2026-09-02-rdiag-shared-dt-angle-headscale-design.md`; and
- `docs/plans/2026-09-02-rdiag-shared-dt-angle-headscale-plan.md`.

The primary implementation and qualification paths are

- `lit_gpt/rdiag_shared_dt_kla.py`;
- `lit_gpt/kla_ops/rdiag_shared_dt.py` and `lit_gpt/kla_ops/rdiag_bounded_phase.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_production_host.py` and `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_training_health_host.py`;
- `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab.py` and `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py`;
- `scripts/train/rdiag_kla/rdiag_recipe.py`, `scripts/tests/rdiag/attest_rdiag_shared_dt_gate.py`, and `scripts/train/rdiag_kla/rdiag_shared_dt_qualification.py`;
- `scripts/tests/rdiag/prepare_rdiag_shared_dt_source_gate.py` and `scripts/tests/rdiag/prepare_rdiag_shared_dt_qualification.py`;
- `scripts/tests/rdiag/rdiag_shared_dt_production_soak.py`, `scripts/tests/rdiag/run_rdiag_shared_dt_production_soak.sbatch`, `scripts/tests/rdiag/attest_rdiag_shared_dt_production_soak.py`, and `scripts/tests/rdiag/attest_rdiag_shared_dt_qualification.py`; and

- `scripts/train/rdiag_kla/submit_rdiag_shared_dt.py`, `scripts/train/rdiag_kla/rdiag_job_control.py`, and `scripts/train/rdiag_kla/rdiag_batch_supervisor.py` for the separately snapshotted production run and its mandatory evaluation tail.

The immutable qualification source is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/
rdiag_shared_dt_auth_v3_qualification_
a76dc7b930ae4d85157391d444c40ef7db14ef86,
```

and the campaign root is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
campaigns/
rdiag_shared_dt_auth_v3_a76dc7b930ae4d85157391d444c40ef7db14ef86.
```

The failed-soak investigation is isolated in sibling worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_soak269_diag_20260903
```

on branch `codex/rdiag-shared-dt-soak269-diagnostic-20260903`, based exactly on `a76dc7b930ae4d85157391d444c40ef7db14ef86`. The principal diagnostic code is `lit_gpt/rdiag_shared_dt_training_health.py`; the retained journal is under `production_soak/output/evidence/qual_rdiag_shared_dt_soak3300_a65918f1771446e7acfbffcc3e5c7dc3`, and the job logs are under `production_soak/logs` within the campaign root. The completed replay plan retained in that diagnostic branch is

```
docs/plans/2026-09-03-rdiag-shared-dt-soak269-diagnostic-plan.md.
```

Its replay entry points are `scripts/analyses/replay_rdiag_shared_dt_soak269.py` and `scripts/analyses/run_rdiag_shared_dt_soak269_replay.sbatch`.

The diagnostic line progressed through initial and early-mode commits

```
f82fa86f52c0f661d0ddbc74197d1dbc82a9e8f2,
28531cf959304ab26b22789f2ea4ce082adbed13,
```

before the final search commit and tree

```
06eaf237f5be9f79ee8d00075b79dcc0e7a9e96a,
f1ecf305edc04824ce14042fe75dbbba106f21c0.
```

The final immutable replay source is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/
rdiag_shared_dt_alpha_zero_search_
06eaf237f5be9f79ee8d00075b79dcc0e7a9e96a,
```

with source-manifest SHA-256

```
81f40cb7ccd697483d7b2757e0c728da18c4ee3a99b85b54be2f63e9af51cffe.
```

Its replay job 1829305 wrote only below `diagnostic_runs/soak269_replay_06eaf237f5be_1829305` and produced neither a qualification result nor a seal.

The then-current qualification checkout, containing the minimal lower-endpoint production-policy correction, is isolated from both the main checkout and diagnostic harness in worktree

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_worktrees/rdiag_shared_dt_alpha_zero_fix_20260903`

on branch `codex/rdiag-shared-dt-alpha-zero-fix-20260903`. It is based exactly on the reviewed head-scaled commit. The production-policy change itself is commit and tree

`46ba8ec3105177a9caba2f2c153e9b71c8b62cac`,
`e38fdab82c68300a361613e4ee3a8c784b7c1a12`.

The executable plan is `docs/plans/2026-09-03-rdiag-shared-dt-alpha-zero-policy-plan.md`. Aside from adding that plan, commit `46ba8ec...` changes only `lit_gpt/rdiag_shared_dt_training_health.py`, its focused host test `scripts/tests/rdiag/test_rdiag_shared_dt_training_health_host.py`, and one existing case in `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`; it does not change model or kernel math.

The first qualification attempt exposed a test-construction issue rather than a kernel failure. Its new subcase forced $DT = 120$ at all 33 tokens, roughly 10.9 times the reproduced maximum, and therefore also imposed phase stress outside the replay-observed range. Immutable job 1829420 passed 108 of 109 tests; the modified full-layer case passed its output comparison and finite elementwise/relative-L_2 hidden-gradient checks, but its hidden-gradient cosine was $0.996079 < 0.999$. The gate failed closed, result SHA-256 was `d64821ee25f44c06aad4b6fb8d27b1433af328381be9a61d08e6e244e0a117bc`, and no seal was created. This failed snapshot and campaign are retained as evidence and are not reused.

Correction commit and tree

`83118d2d96325c9b9b555d60ad4316c950c5a12d`,
`8c05ae8f2b5b205a17ba64042c4fad337a1d4d45`

restore the ordinary strict output/all-VJP oracle case unchanged. A separate subrun within the same catalog item uses replay-bounded $DT = 6.53125$ and rate 16, giving exponent 104.5 and exact FP32 $\alpha = 0$. It proves that Kappa receives raw zero, S3 receives the finite BT32 prefix of $\log_2(10^{-6})$, the full forward matches the literal oracle, all model gradients remain finite, and live Kappa/S3 cotangents are finite; no tolerance or 109-test inventory entry changed. The related host sweep passed 131 tests. Development H200 job 1829576 passed the corrected item in 117.26 seconds; its stdout SHA-256 is `271d1ceb10772e4516cb9fe897e67cb7989e9af07939e329120a3d394f36f3a2`. Two independent exact-SHA reviews approved this final commit without blockers, and the branch is published.

The superseded 4228-file qualification snapshot for `46ba8ec...` is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/`
`rdiag_shared_dt_auth_v3_qualification_`
`46ba8ec3105177a9caba2f2c153e9b71c8b62cac`,

with source-manifest SHA-256

`2507dede155e9e50df876a493170d8ccfc2523413bab115e7a3f0e3369f21b93`.

Failed campaign `2f02ccb07eac4837bd6416546b17112c` is rooted at

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
campaigns/
rdiag_shared_dt_auth_v3_
46ba8ec3105177a9caba2f2c153e9b71c8b62cac.

Compute-side preparation job 1829417 passed after the login namespace correctly rejected a distinct live-data inode identity. Job 1829420 ran from execution-manifest SHA-256 b23ea908ac550bc14c76267ff8e295e06f620d21412e37c766787f8ecfd72eb2.

The now-superseded immutable qualification manifest for the corrected lower-endpoint test covers 4228 source files at

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/
rdiag_shared_dt_auth_v3_qualification_
83118d2d96325c9b9b555d60ad4316c950c5a12d,

with source-manifest SHA-256

52a82f97849f400730dfa2496472017916ca84e36dd785a7bac9e2da629936df.

That now-superseded campaign, a262363313d141e9bda423b0ce243940, is rooted at

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
campaigns/
rdiag_shared_dt_auth_v3_
83118d2d96325c9b9b555d60ad4316c950c5a12d.

Compute-side preparation job 1829624 passed. Production-H200 gate job 1829629 then completed on H200 node h200-028-028 with scheduler exit 0:0 and all 109 tests passing. It ran from execution-manifest SHA-256 98488ffdb2db208f015e34df678ef57a4003cc8ac8c29337aa7648755509e0df. Its retained result has SHA-256 32c5b2eed517f2add1e327d841c43f012858454edcf975bb5681403ec45e8675; the JUnit report and run log have SHA-256 values 507bfa8de347b1967d573b41578c3c3760a23b42e8b412f4076fe7b443dcba99 and c6357f6d46f284dda75280ae94c53d3668e0e78bdce92abd6b5c051e83fed30d, respectively. Compute-side attestor job 1829796 completed successfully and published the read-only H200 bootstrap seal with SHA-256 6f900869b3c13e6ef41e8920a52f9115bc31f7e69f7f8169a8b4014dc55b05f8. Five compute-side preparers then authenticated that seal and emitted read-only execution manifests for train-smoke, checkpoint-fault, authenticated-resume, gradient A/B, and the 3300-update soak, with SHA-256 values respectively

0d1e63001f57615c21a494e11eacfda8f04f464d258eb6d3c10207557f70fdc8,
9287308502d9e2bbffc6ff0bb2e71506c4ea9d14758a703edbef8519914157d2,
34280cd404e1680b3bc13a94bea26678cd4b24fd37bc49c4d64ad5be817ed281,
8f4fd8e74f56c3770c0464a385faac255561933b510d741d3820ddcae4ef7da7,
5f3e7a352e529666aebfbbd841c06413310b7db0a00525f45184880772e0f8ed.

Only their emitted commands were submitted, as jobs 1829974–1829978. Train-smoke, checkpoint-fault, authenticated-resume, and gradient A/B jobs 1829974–1829977 passed and were independently attested. Their seal SHA-256 values are respectively

472ebcf70fa24a50870b68b25024c74df296d6f35f1e218eeeb36965458dd13a,
b4f44aa38f2e11397f300f88f50b03881008ebe4d19ea7afa688ce9565ea416f,
153c83d8be341bf7600990adeea69bb35e792b441459159e6fdc928316036a12,
164138bf3a3c2809d8ed4fb7dd93a7a829fa42714fd123ddd8a634b6e3bc2d11.

Gradient A/B again passed with shared/no-rotation median and 95th-percentile ratios 1.179 and 1.260. Production-soak job 1829978 ran on nodes `h200-207-042` and `h200-211-043`; its immutable-source and runtime-identity stages completed successfully. It crossed the original failure point: updates 268, 269, and 270 all completed with zero health failures. At update 269, loss was 5.3710, global gradient norm was 1.2287, and 11211776 alpha elements were below the S3 floor but remained diagnostic-only. The run then completed 675 healthy updates. Its final retained record has loss 4.0055, global gradient norm 1.4537, and zero health failures.

The next pre-optimizer check, boundary 676, failed closed with aggregate failure count 256 on all 16 ranks; update 676 was not applied. The Python step exited 1 and the surrounding Slurm job terminated with exit `15:0`. The ordinary soak evidence deliberately does not retain the failed-key breakdown, so at that stage the responsible predicate was not identified. No result or seal was emitted. The 675-line update journal, stdout, and stderr have SHA-256 values respectively

```
d26a63a89430a738c689d3e1fddc2d554ce30f3c400b1258c63c2a3fd2ef31cc,
958939cf35074c8f9ca8611a19ec3dabd2830bd3090ee2e2bb104a12627dfd93,
202e1bf22ac711421f006ef01c4c9820ac1522726cb4569480f635e7217c5a28.
```

Boundary-676 diagnostic replay. The non-authorizing investigation is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_boundary676_diag_20260903
```

on branch `codex/rdiag-shared-dt-boundary676-diagnostic-20260903`. Its executable plan is `docs/plans/2026-09-03-rdiag-shared-dt-boundary676-diagnostic-plan.md`. Diagnostic-only health rendering was introduced at commit `ef4ee3c053bc1d34243f4086edd88e62b85f73d4`; the reviewed replay harness is commit and tree

```
a9ce0db738dc65c3d72e4bcd85b955815952b163,
bfb397412c1977acaef4742850738f69e948fc87.
```

The relevant code paths are `lit_gpt/rdiag_shared_dt_training_health.py`, `scripts/analyses/replay_rdiag_shared_dt_boundary676.py`, `scripts/analyses/run_rdiag_shared_dt_boundary676_replay.sbatch`, and `scripts/tests/rdiag/test_rdiag_shared_dt_boundary676_replay_host.py`. The focused replay suite passed 29 tests, the broad shared-DT host sweep passed 522 tests with four skips, and three independent reviews approved the exact staged tree that became the replay commit.

The immutable 4232-file diagnostic source is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_boundary676_diagnostic_
a9ce0db738dc65c3d72e4bcd85b955815952b163,
```

with source-manifest SHA-256

```
6f7e52aa691c684543be51aeda0b6ab3b8d5d43c90b753b306070af68209da45.
```

Replay job 1830570 ran from that snapshot with the pinned failed-run data manifest and the exact original nodes `h200-207-042,h200-211-043`. It completed with scheduler exit `0:0`; every retained record carries `authorizing:false`, and no qualification result or seal was emitted. The replay stopped naturally before optimizer boundary 649 after 648 completed updates. Its sole fatal key was `alpha_at_or_above_one_count=640`, exactly five shared scalar sites expanded over width 128.

Global `alpha_max` was exactly one. All DT, $A_{\log}$, $\exp(A_{\log})$, and decay-exponent values were finite; $DT_{\min} = 1.1172518910 \times 10^{-8}$ and the smallest positive decay exponent was $3.4472314780 \times 10^{-8}$. Parameter and gradient nonfinite counts were zero. Thus the captured failure is rejection of the finite FP32 identity-decay endpoint, and no monitored nonfinite predicate fired. The replay alone is not a Kappa/S3 oracle check; the later targeted $K = 128, T = 33$ H200 subrun supplies that forward/VJP evidence. The observed exponent is slightly above the ideal round-to-nearest-even crossover $-\log(1 - 2^{-25}) = 2.9802322832 \times 10^{-8}$; CUDA's transcendental implementation need not be correctly rounded, so the retained output value and bitwise endpoint test, rather than the ideal crossover alone, govern the implementation claim.

The replay is deliberately classified `DIFFERENT_FAILURE`: it did not reproduce the old trajectory, boundary, or count; instead it failed at boundary 649 with `alpha_at_or_above_one_count=640`, while the original boundary-676 key remains unknown. Across the 648 comparable updates, maximum absolute loss and gradient-norm differences were 0.1493 and 1.3539. This evidence does not identify the old boundary-676 key retroactively, but it independently demonstrates that the replayed `83118d2...` upper-endpoint policy rejected a valid production-shaped state. The retained artifact root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/boundary676_replay_a9ce0db738dc_1830570`.

The summary, trajectory, invocation, inner stdout, inner stderr, and artifact-inventory SHA-256 values are respectively

`9a8c357b37731924aed61e02b7bf8ea599b943f7508cb93f91fb7bd572c89157`,
`7eb3ab52f083079f85059522c7b0a219e22cacdd3923c3b55df135d59b851ac7`,
`82848cb2fb25f4b448cebb2e87dddf197d320c7c962dbc6946e654ce6a531fa7`,
`859d3c15b6223acafabddecc491afd896ebaf22368b5fc54823af481a05c18f4`,
`645326b62f8847a96dcc47cb8ddf9b39ed214dde65d6b731131c3f8b90e8b01c`,
`af23c4fef6684ae97102f3d81bb4cb6edc46e116623cc5e9f23fca88da78e370`.

All 24 inventoried artifact hashes independently revalidated. The terminal outer scheduler log and empty scheduler error file have SHA-256 values respectively

`2034750abdc9f1baea2074a0a8a28f2e6196838a1beac4fa84ad984d9ddbc151`,
`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`.

The resulting narrow correction is isolated from both qualification and replay evidence in worktree

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_worktrees/rdiag_shared_dt_alpha_one_fix_20260903`

on branch `codex/rdiag-shared-dt-alpha-one-fix-20260903`, based exactly on `83118d2...`. Its plan is `docs/plans/2026-09-03-rdiag-shared-dt-alpha-one-policy-plan.md`. The reviewed and published correction commit and tree are

`e5d07ad0f49f74e90d89a0b49ab8279b39152546`,
`2573a97d33244e9125ba94557b428868816360f4`.

That commit changes exactly the plan and three implementation/test files

- `docs/plans/2026-09-03-rdiag-shared-dt-alpha-one-policy-plan.md`;

- `lit_gpt/rdiag_shared_dt_training_health.py`;
- `scripts/tests/rdiag/test_rdiag_shared_dt_training_health_host.py`; and
- `scripts/tests/rdiag/test_rdiag_shared_dt_production_h200_v1.py`.

It retains `alpha_at_or_above_one_count` as telemetry, adds `alpha_above_one_count`, and makes only strict $\alpha > 1$ fatal; negative and nonfinite checks are unchanged. The 109-case H200 catalog gains no test node: an existing full-layer case now includes a positive-exponent exact-one subrun covering raw FP32 bits, Kappa, the BT32 S3 prefix, literal recurrence agreement, and live finite VJPs. Host policy tests passed 36/36, the broad host sweep on the final pre-commit worktree passed 492 tests with four skips, and targeted H200 job 1830711 passed in 114.16 seconds. This was non-authorizing mutable-worktree development evidence, run before the final commit; the immutable full H200 gate is job 1830943 below. The targeted job's stdout and stderr SHA-256 values are

`689c9451b67e5e9593fb15bc858723f4d4e7e89d3b47fdb90ebbce5044501442`,
`57ac2d0201a2657267a97ef3371038e486fe85eac5d21983843138591659f847`.

Three independent exact-SHA reviews approved the correction without blockers.

The superseded `e5d07ad...` source snapshot manifest covers 4229 source files; the snapshot directory also contains `SOURCE_MANIFEST.json`. That snapshot is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/rdiag_shared_dt_auth_v3_qualification_`
`e5d07ad0f49f74e90d89a0b49ab8279b39152546`,

with source-manifest SHA-256

`3f6a8ae9176ddb9e9aee323febf6cb5d9dc22bc88d8933540be727c23a604832`.

Its corresponding campaign `7a24d88d497d4ad28b572b4ee885ac90` is closed and incomplete; it is rooted at

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`campaigns/rdiag_shared_dt_auth_v3_`
`e5d07ad0f49f74e90d89a0b49ab8279b39152546`.

Compute-side preparation job 1830934 passed and emitted production-H200 execution-manifest SHA-256 `0d9277ab08b69f8d9e673e96cdd2d91ce8c753ce1d495879850edf0355377426`. Its exact control path is

`production_h200/control/`
`production_h200_7a24d88d497d4ad28b572b4ee885ac90.manifest.json`

relative to the campaign root above. The corresponding scheduler logs are under `production_h200/logs/`, and retained gate artifacts are written below `production_h200/artifacts/production_h200_7a24d88d497d4ad28b572b4ee885ac90/`. The resulting H200 gate, job 1830943 on `h200-004-175`, completed with scheduler exit `0:0` and 32:21 wall time; pytest reported all 109 cases passed in 405.96 seconds after the source/runtime preflight. Its result, JUnit report, and run log have SHA-256 values respectively

`6ac6a4ce2775230cfcfd929aa77638c968ca6841b4a863a85824e479019f67da`,
`b6132c80118ec46578178ab567cae9a76361d30d8476d403b8071a3c154b65be`,
`5301d528c8b215fddca189209c17f8f3efefe7cbd831204e141ec29aed067c41`.

Its outer stdout has SHA-256 `e33b84600aa1d7896723fcac68d40d342c5264914c73a052b1c608863d303f2f`; the empty stderr has SHA-256 `e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`.

The numerical result cannot authorize downstream work because its retained `run.log` is the exact 2706-byte prefix of the 2728-byte scheduler stdout: only the final 22-byte `PASS:production_h200` line is absent. At `e5d07ad...`, both source-gate runners redirected through an asynchronous process-substitution `tee`, froze `run.log`, printed PASS, and exited without waiting for that writer. The race affected both production-H200 and train-smoke; the attestor's requirement that both closed logs contain PASS was correct and remains unchanged.

The first attestor allocation, job 1830944, had an operationally insufficient 20-minute wall limit. After 5:43 it had read 34496.23 MiB of the runtime closure; the corresponding prior successful attestor read 270926.75 MiB in 42:47, projecting this attempt well beyond its limit. It was therefore manually cancelled before publishing a seal; dependent preparer 1830946 never allocated and was also cancelled. Replacement attestor job 1831239 used the byte-identical command and evidence pins with a one-hour limit. It completed the full closure read and then correctly failed `2:0` after 57:28 on the missing retained-log marker; it emitted no seal. Dependent preparer 1831240 never allocated. A redundant fail-safe attestor, job 1831513, was cancelled after the deterministic artifact defect was identified rather than spending another full validation cycle.

The logging correction is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_source_gate_log_fix_20260903
```

on branch `codex/rdiag-shared-dt-source-gate-log-fix-20260903`, based exactly on `e5d07ad....` Its executable plan is

```
docs/plans/
2026-09-03-rdiag-shared-dt-source-gate-log-finalization-plan.md.
```

Working-location index. The worktree above is the historical qualification-source checkout, on branch `codex/rdiag-shared-dt-source-gate-log-fix-20260903` at the exact reviewed commit `1fce1b9...`; current diagnostic and control development is isolated in the generation-zero, forward-determinism, RMSNorm-control, trigger-capture, and matched-intervention worktrees named below. The main checkout is not the source of any qualification job. Within the historical source worktree, the shortest paths into the implementation and its controls are:

- `lit_gpt/rdiag_shared_dt_kla.py` for the layer parameterization and the $H^{-1/2}$ angle head scale;
- `lit_gpt/kla_ops/rdiag_shared_dt.py` and `lit_gpt/kla_ops/rdiag_shared_dt_reference.py` for the composed GPU path and reference recurrence, with phase/clock helpers beside them;
- `lit_gpt/rdiag_shared_dt_training_health.py` for the alpha endpoint telemetry and fatal-domain policy;
- `scripts/tests/rdiag/` for H200 cases, distributed smoke/soak/fault gates, runners, and attestors; and
- `scripts/train/rdiag_kla/` for recipe pinning, campaign/job control, authorization, and the production submitter.

The design and implementation plans for the numerical correction are, relative to that worktree,

- docs/plans/
  2026-09-02-rdiag-shared-dt-angle-headscale-design.md; and
- docs/plans/
  2026-09-02-rdiag-shared-dt-angle-headscale-plan.md.

The two subsequently discovered finite-precision endpoint policies are recorded in the corresponding alpha-zero-policy and alpha-one-policy plans dated 2026-09-03. The boundary-269 and boundary-676 plans live only in their respective historical diagnostic worktrees. Those filesystem-writable worktrees are retained for investigation but are not qualification sources; the current immutable source snapshot and campaign root are named below.

The only intended runtime changes are scripts/tests/rdiag/run_rdiag_shared_dt_production_h200_v1.sbatch and scripts/tests/rdiag/run_rdiag_shared_dt_train_smoke.sbatch; regression coverage is in scripts/tests/rdiag/test_rdiag_shared_dt_source_gate_host.py. The corrected lifecycle is PASS emission, closure of both pipe writers, an explicit wait for the captured tee PID, and only then read-only log publication. It preserves any earlier nonzero status and makes a logger or chmod failure fatal. No result, seal, aggregate, or attestor schema is weakened.

The reviewed and published logging correction is commit/tree

```
1fce1b9a8e4a80d4da82a689dfb69983109aad0c,
297b4b1f3f4ba7e4ac64ac35df0510c9bdea69be.
```

It changes exactly the two runners and the source-gate host test named above, plus the tracked plan. A focused selection over test_rdiag_shared_dt_source_gate_host.py, test_rdiag_shared_dt_production_h200_v1_host.py, test_rdiag_shared_dt_train_smoke_host.py, and test_rdiag_shared_dt_gate_runners_host.py passed 134 tests. The full 22-file RDiag host sweep passed all 823 tests, and independent collection still reports exactly 109 H200 cases. Two earlier broad attempts were deliberately excluded: concurrent test artifacts filled the 512 MiB /var/tmp filesystem in one, while placing temporary trees on Lustre changed the filesystem-specific test branch in the other. The final sweep used a spacious local filesystem and had no failures. Three independent exact-SHA reviews approved the commit without blockers.

The replacement immutable source snapshot contains 4230 tracked regular files and excludes only the approved paper gitlink. It is

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_auth_v3_qualification_
1fce1b9a8e4a80d4da82a689dfb69983109aad0c,
```

with source-manifest SHA-256

```
429348ceb16350782bc0eb2b4a34affabc5eb73f1d3ac997e5da42a2d1949067.
```

Fresh campaign ae2041e3d1fd4b1ca7c1d609184c2642 is rooted at

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
campaigns/rdiag_shared_dt_auth_v3_
1fce1b9a8e4a80d4da82a689dfb69983109aad0c.
```

Compute-side preparation job 1832038 completed with execution-manifest SHA-256

```
a1759ea8f8b058b4ab49b235d7ce0d083d7760cc41447d0d9447e01515ee925e.
```

The manifest is

`production_h200/control/`
`production_h200_ae2041e3d1fd4b1ca7c1d609184c2642.manifest.json`

relative to that campaign root. Its authenticated emitted command was checked as an exact argv sequence inside H200 allocation 1832039 and submitted as job 1832040. That gate completed `0:0` on `h200-052-125` in 31:54; all 109 cases passed in 423.33 seconds. The result, JUnit report, and retained log have SHA-256 values respectively

`e39ebe47ce0e1bd43f23c312c75269a5a6300b6f367f6f43df7906568bc8d057`,
`a654c778be9062d4cc9f41b62b0f779732f09bf0c5e9c27521538384c1c9a732`,
`5335a009a8407fcf2965cc67eb09f02124d562d378c98019e05119cf57e1401d`.

The mode-0400 retained log is byte-identical to scheduler stdout, includes exactly one terminal `PASS:production_h200` marker, and the scheduler error file is empty (SHA-256 `e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`). This directly exercises and resolves the former publication race. The two-hour external attestor is job 1832042; it completed `0:0` in 52:16 and published a mode-0400 bootstrap seal with SHA-256

`a6c63dae54416ad4085631de5e7d27ee97423ad881a5d1f75596d68d4dc35cff`.

Five manifest-only preparation jobs were dependency-bound to that attestor: production soak 1832100, train smoke 1832101, checkpoint fault 1832102, authenticated resume smoke 1832103, and gradient A/B 1832104. All five completed `0:0`; their stderr logs are empty and their mode-0400 execution-manifest SHA-256 values, in that order, are

`2039a4b702586e5549c6a92bcfb0d0b3054151df8af290086e153439674c9873`,
`36c6b0a2597dd6ee31d95b048d9434eb6a021f2f8a65a26736fa4fca6822e0df`,
`f2ae3776688b6a3c06e6fd7b2f6e1e71b303e50663fb2b296bebf799797fb63f`,
`5ca194e4c54c7342c498689000d377acbf283e1e5a0be9075e7b9116c6e9a15c`,
`db126e731f9bae7cdd1172fa4815a966dc49084ecb21310ca0026dfa30e5d119`.

Control allocation 1832771 independently rehashed each manifest, loaded it through the pinned control code, rebuilt the scheduler argv inside an H200 allocation, required exact equality with the preparer-emitted argv, and found no existing job with any of the five unique scheduler comments. This compute-side step was required because the login node intentionally has a different data-root device identity. It submitted the exact workloads and dependency-bound external attestors as the following pairs:

`production-soak1832774->1832775`,
`train-smoke1832776->1832777`,
`checkpoint-fault1832778->1832779`,
`authenticated-resume1832780->1832781`,
`gradient-A/B1832782->1832783`.

The four short workloads and attestors then completed successfully:

- train-smoke jobs `1832776->1832777`, seal SHA-256 `d04cbec0b679496a8d4e05c22435c919ce1dbce455ea1e08dd48bbc04abad556`;
- checkpoint-fault jobs `1832778->1832779`, seal SHA-256 `d382d2ef31dbb49588316a7850f8c022ee3796681ee6aa418e10a53f0c0902aa`;

- authenticated-resume jobs `1832780->1832781`, seal SHA-256 `dc1ab93029eca699c27d4d1a8ac23a946cc6ff2c76c95d30de77edd82d897e23`; and
- gradient-A/B jobs `1832782->1832783`, seal SHA-256 `a17fb2ecc2f116c41c7b333853571a8665b4811a23685b8530fea071b1e12e2b`.

All eight scheduler exits are `0:0`; every external-attestor stderr is empty, and each independently rehashed seal is an owner-held, single-link, mode-0400 regular file. The new train-smoke retained log is byte-identical to scheduler stdout and contains exactly one terminal PASS marker, providing a second end-to-end exercise of the log-finalization fix. Checkpoint-fault and authenticated-resume emitted the same framework initialization, FSDP deprecation, and process-group shutdown warnings seen in the earlier successful campaign; neither emitted a Python traceback, and the external attestors accepted their exact results.

The two-node soak allocation 1832774 started at 2026-09-03 20:15:33 UTC on `h200-232-233` and `h200-233-046`. Slurm emitted two bootstrap warnings because it probed the exported job-private `TMPDIR` before the first per-node bootstrap step had created that directory, and therefore used `/tmp` for that step; the runner then created and used its intended mode-0700 cache trees. This is retained scheduler stderr, not the training failure.

The journal crossed both historical failure boundaries with zero aggregate health failures. Its exact update-269 row records loss 5.395138889551163, global gradient norm 1.213560938835144, and 26,092,800 alpha values below the diagnostic S3 floor out of 17,179,869,184. Its update-676 row records loss 4.053754523396492, global gradient norm 1.0578112602233887, and 102,754,304 below-floor values out of the same denominator. The workload then failed closed at the pre-optimizer boundary for update 790. All 16 ranks reported `SoakEvidenceError:globalgradientexceedsthesoaklimit`; job 1832774 ended `FAILED1:0`, and dependency-bound attestor 1832775 was cancelled without allocation. No soak result, final checkpoint, seal, or aggregate exists.

The retained journal contains exactly 789 contiguous accepted rows and has SHA-256 `4814e6c0cedbdc4efdceefb4b944ca2e8b1df357d4f7a49118857a64f8da286f`. Its median and lower-empirical 95th and 99th percentile pre-clip norms are respectively 1.14646, 2.27364, and 3.35581. The maximum is 7.970221519470215 at update 789; excluding that row, the maximum is 4.06596 at update 24. Updates 783–789 measure 3.39357, 1.82029, 1.61033, 2.27467, 2.26284, 3.04553, and 7.97022, followed by the unretained update-790 rejection. Loss at the rejected boundary was finite, and every accepted row has zero health failures.

The name `after_clip` is temporal rather than semantic. Lightning delegates this call to FSDP, which returns the accumulated distributed norm *before* multiplying gradients in place by approximately `min(1,1/(norm+1e-6))`. Thus update 789 was subsequently clipped to approximately one; update 790 had a rank-equal pre-clip norm that was nonfinite, negative, or above 10 and was rejected before `optimizer.step`. Its exact scalar and predicate branch are irrecoverable from this run: the harness checks the combined predicate before constructing or appending a record, and named-gradient sampling begins only at update 3000. The run retained only its generation-zero checkpoint.

This is a genuine failure of the frozen qualification contract, but not yet proof that the 750M model is unstable. The limit 10 was copied from the older bounded-E2 soak. That accepted E2 screen observed a sampled maximum 1.888 over updates 300–500, but it did not calibrate the tail of a completed 3300-update shared-DT run. Conversely, the separate 220M gradient-v3 gate uses 100 only as a gross fail-fast ceiling around a matched shared/no-rotation ratio experiment; it cannot justify changing the 750M soak threshold to 100. The late rise makes attribution mandatory before either a model change or a versioned gate-policy change.

The non-authorizing investigation is isolated in worktree

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_worktrees/rdiag_shared_dt_soak790_diag_20260903`

on branch `codex/rdiag-shared-dt-soak790-diagnostic-20260903`, branched from `1fce1b9...`. The reviewed and published execution commit is `47bb7812da1bd5809d974ea9c5504209f3a357f3`, tree `2fc10ee5320a418a72960a86145083916a3d739c`. Post-execution evidence hardening is the current branch head:

commit `4091b2572f19f8cefaab8932b2d035969f663ec0`,
tree `bf8abd733df27ba03331fadf35560107fadf1e8b`.

It does not alter the executed source. Relative to that worktree, the completed replay plan is `docs/plans/2026-09-03-rdiag-shared-dt-soak790-diagnostic-plan.md`; its entry points are `scripts/analyses/replay_rdiag_shared_dt_soak790.py` and `scripts/analyses/run_rdiag_shared_dt_soak790_replay.sbatch`. The replay keeps the production recipe, ordered data, seed, optimizer schedule, 2-by-8 topology, FP32 FSDP buffers, clipping, and strict limit under measurement-only instrumentation. This is a same-recipe replay, not bit-exact production reproduction: read-only scans, copies, and reductions can perturb GPU scheduling, and "same boundary" denotes only the boundary number. It revalidates the failed run's complete runtime fingerprint, computes per-rank target-token hashes and pre-clip category and top-parameter diagnostics at every boundary, and retains the hashes for a gradient-rejection boundary. It stops non-authorizingly on a natural production-health failure, a finite norm above 10, a nonfinite returned norm or post-health named gradient, or exhaustion through update 820. Nonfinite values use canonical JSON-safe tokens, finite-only norms are labeled as such, and every offending parameter remains separately listed. The validator requires the exact 32-field PRE-health inventory, the full-update alpha denominator 17,179,869,184, and name-to-category agreement for every retained contributor. Physical-family nonfinites remain owned by the earlier production-health failure path.

The final pre-execution replay, health, and pretrain-integration host suite passed 87 tests. Independent runner, schema, and mathematical-contract reviews approved execution commit `47bb7812...`. A Git-object snapshot was materialized at

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/rdiag_shared_dt_soak790_diag_47bb7812da1bd5809d974ea9c5504209f3a357f3`

with source-manifest SHA-256 `a948eb8a0540a2ff7625f344372a2d04389865bb5e3a977ad7ceff123e4a742f`. The manifest closes 4,234 tracked regular files; its `SOURCE_MANIFEST.json` is stored at the snapshot root. Two-node H200 replay job `1833926` ran on `h200-101-019` and `h200-208-076` from 2026-09-03 22:40:57 through 23:02:14 UTC. It and all Slurm steps completed `0:0`. The rank-complete result is `NO_GRADIENT_LIMIT_EXCEEDANCE_THROUGH_820`: all 820 returned norms were finite, nonnegative, and no larger than 10; all named-gradient nonfinite counts and all health-failure counts were zero. Update 790 measured norm 1.0757828950881958 and loss 4.024734705686569, update 820 measured norm 4.972261428833008, and the replay maximum was 7.394214630126953 at update 817. Thus the replay did not reproduce the original update-790 rejection.

The replay is not a matched trajectory. Of the 789 comparable records, 673 differ in at least one shared field; floating-point norms differ at update 1, alpha-below-floor counts first differ at update 3, and every update from 139 through 789 differs. Original versus replay update 789 is respectively

7.970221519470215 versus 1.4658762216567993. Gradient norms over 1–789 have Pearson correlation 0.5998 and first-difference correlation 0.0389; over 1–99 those values are 0.999565 and 0.99191, while update 100 starts the first five-update run with absolute norm differences above 0.25. Loss remains tightly same-step coupled (Pearson 0.999511), showing the same broad recipe but not the same numerical state. No original update-790 input hashes exist, and this no-rejection outcome retained no replay batch hashes, so data equality at that boundary is unproved.

The late replay excursion is broad rather than an isolated radial failure. Across updates 783–820, squared gradient-norm mass is 45.57% ordinary/backbone, 32.91% angle, 21.50% DT weight, 0.026% DT bias, and 0.00085% `A_log`. At the maximum, update 817, category norms are 5.2138 ordinary/backbone, 4.5117 angle, 2.6695 DT weight, 0.0909 DT bias, and 0.0271 `A_log`. This localizes co-amplification but is not a causal decomposition; it supports neither removing rotation nor changing the frozen limit.

The finalized artifacts are under

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/soak790_replay_47bb7812da1b_1833926/artifacts`.

The retained evidence digests are:

summary: `9620387f55d38bfde32e5763b1ecf72c1c4fe23223ea6d01d051acb246a718f5`,
trajectory: `d8fbb3c1fdea2bc1f63f089cb457b9ee832372cd828465fbee2070327801a7fc`,
invocation: `ff2a6a83950942e2d7ba12d8ac1bac163264e628f6d34a97e010ab27ad644634`.

An independent audit found one packaging defect: the invocation named Slurm's ephemeral spool copy of the runner, which disappeared after cleanup. Its recorded SHA-256 and size exactly match the immutable snapshot runner, and substituting that path regenerates the summary byte-for-byte, so the result is corroborated but is not self-validating as originally packaged. Commit `4091b2572f19f8cefaab8932b2d035969f663ec0` fixes future packages by retaining a mode-0400 artifact-local `runner.sbatch`, binding it to the retained source-manifest entry, and rejecting external or writable runner records. Its focused suite passes 48 tests. There is no evidential reason to rerun job 1833926 merely to repair this historical packaging defect.

The shortest code paths for this boundary are `pretrain.py` (data fetch, health, clip, optimizer, and checkpoint lifecycle), `data.py` (stateful worker/buffer order), `lit_gpt/rdiag_checkpoint.py` and `lit_gpt/fsdp_strategy.py` (full model/optimizer and per-rank sidecars), and `scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py` (the existing connected-zero rotation intervention). Replay-specific instrumentation remains confined to the helper and runner named above.

Remaining work. The head scale and gradient-v3 closure remain frozen and published at `a76dc7b930ae4d85157391d444c40ef7db14ef86`; the alpha-zero policy source is frozen and published at `83118d2d96325c9b9b555d60ad4316c950c5a12d`. That source crossed the former boundary-269 failure but did not pass the full soak. The alpha-one correction is frozen at `e5d07ad0f49f74e90d89a0b49ab8279b39152546`; the failed source head is its log-finalization child `1fce1b9a8e4a80d4da82a689dfb69983109aad0c`. Every existing seal is source-bound historical evidence; the five successful non-soak seals cannot authorize a source or policy change.

The formerly proposed production root for `1fce1b9...` remains deliberately absent:

`/lustre-storage/checkpoints/ngocbh/hdn/production/`
`rdiag_shared_dt_auth_v2_1fce1b9a8e4a80d4da82a689dfb69983109aad0c`.

It will not be created from a failed qualification. A future production root, if authorized from a later source commit, will have fixed children `source_snapshot`, `manifests`, `control`, `output`, and `logs`. The separate production snapshot must be created atomically by the immutable launcher's `prepare-fresh` mode; the qualification snapshot is not reused as the training snapshot. The four recipe digests listed below are historical pins for `1fce1b9...`, not reusable authorization for changed source:

- `rdiag_kla_shared_dt_v1_220M`: `8f4d2d21f38ee93bda045e74d80bf6c5c1dd1feef9102e3ab413583b1542d9e1`;
- `rdiag_kla_shared_dt_v1_750M`: `7a07b7af54a9df20736fc85c2eb615900c969c8a977ca12836cee210f301b2c4`;
- `rdiag_kla_shared_dt_v1_1.3B`: `f11e601a6a60fe67dd069ec5fc72954c2f32396bda4d910a32b13ed34f377834`; and
- `swa_rdiag_kla_shared_dt_v1_1.3B`: `225f1cd89233f0a290c527ad5c59ce4b71f9d884dfcdb1a6569044a93bd96b3f`.

The launcher credential prerequisite has been materialized through the no-argument/no-stdout credential path at `/storage/home/ngocbh/.wandb-key`; it is an owner-controlled, single-link regular file with mode 0400. No separate production-training snapshot, execution manifest, output directory, or training job has been created. Because update 790 was not reproduced, the next investigation first locates the earliest divergence from the exact original generation-zero checkpoint. It is isolated in worktree

```
/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_gen0_repro_diag_20260903
```

on branch `codex/rdiag-shared-dt-gen0-repro-diagnostic-20260903`. Its executable plan is `docs/plans/2026-09-03-rdiag-shared-dt-gen0-reproducibility-plan.md`. It pins the original 2.9-GB generation-zero model with SHA-256

```
8a37e667678d2743c5bdcfb3b808c909f980c50b906e2ff9e97e29b02074b581
```

and all 16 rank sidecars. It runs three six-microbatch loader controls: no snapshot (the 1833926 path), snapshot-and-continue on the same loader (the failed fresh-production path), and new-loader sidecar restore. Four common-anchor training arms run in counterbalanced `minimal-A1/instrumented-B1/instrumented-B2/minimal-A2` order for three updates on the same allocation and GPU mapping. Complete local parameter/optimizer Merkle inventories and exact input hashes are prerequisites. Historical process RNG was not checkpointed, so this test localizes only the first observed divergence among captured stages; it cannot alone identify the unique historical cause. This experiment is diagnostic only. The generation manifest and its sidecar/ordered-file composites are pinned respectively to

```
6649a1a68f83b80df9f5600864dc52e51639d19b42a827b78c9eeacef9e9f8f6,
162ff05e5940a427504c0075c82002aa8894091a8bd4f1c22c5bb5e0cb69172f,
6a184a861f445a0eb9ed383e029d726f9cec78f97285df3c350fda95bc52fd86.
```

The validator requires exact zero counters, modes, identities, and the 17-file model-plus-sidecar closure before any arm runs.

Implementation and review are complete in the worktree above. The reviewed execution identity is

commit `4804be477a09f19adbd88d5ca773be721608d9f3`,
tree `b3d882ee859038366f6d18cbd612ad91aa2d31d3`.

It is based on diagnostic-evidence head `4091b2572f19f8cefaab8932b2d035969f663ec0`. Relative to the worktree, its executable entry points and host contract are

- `scripts/analyses/reproduce_rdiag_shared_dt_gen0.py`;
- `scripts/analyses/run_rdiag_shared_dt_gen0_repro.sbatch`; and
- `scripts/tests/rdiag/test_rdiag_shared_dt_gen0_repro_host.py`.

Pre-execution review found and corrected four material harness defects: the post-run snapshot verifier omitted its required private-copy argument; the terminal completion marker preceded verified node-local cleanup; an instrumented-arm wrapper discarded ordinary loss capture; and the classifier could ignore some global-loss, learning-rate, named-gradient, rank-agreement, and loader-to-training input discrepancies. The corrected helper now rederives every field used for classification, binds the restored L2 input matrix to all training arms, requires finite rank-equal pre-clip norms, distinguishes minimal- and instrumented-pair collective differences, and validates an exact 214-file artifact closure before completion. The exact reviewed helper, runner, and test SHA-256 values are respectively

`37b0e2061a145b2b1935a74a362235a1d462457119ed42e4e1efd3d36b7be7bc`,
`f9d5c8a04f7384b360eb04435c32d0a2abeb177e6f6534e91f4910e5b7c1bb14`,
`750d0bae60ca24c7ae7fecffc0a52c2decba915854b5f282910b038bd088bfdc`.

The focused generation-zero suite plus the retained soak-replay suite passes 94 tests; Python compilation, batch-shell parsing, staged whitespace checks, and a fresh exact validation of all 17 generation-zero inputs also pass. Independent artifact/runner, mathematical-contract, and Lightning/FSDP runtime reviews found no remaining Critical or Important issue on these exact bytes. The broader RDiag host sweep first reached 898 passes before 19 late fixture writes failed because the login node's 1-GB `/tmp` filled; these were `ENOSPC` errors rather than assertion failures. Re-running both affected files with a diagnostic-Lustre temporary root passed 82/82, closing all 917 distinct broad-suite tests. The two disposable test-owned temporary trees were then removed.

Commit `4804be477...` is pushed on the branch named above. Its immutable Git-object snapshot is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/`
`rdiag_shared_dt_gen0_repro_diag_4804be477a09f19adbd88d5ca773be721608d9f3`.

The snapshot contains 4,238 tracked regular files, excludes only the external `paper` gitlink, and was independently revalidated against source-manifest SHA-256

`1bbf00a65b30ffd26bd01926e09b7c306c2836054f40d83aab9963904e16c893`.

Two-node H200 diagnostic job `1834859` ran directly from that snapshot on `h200-091-173` and `h200-101-011` from 2026-09-04 01:23:02 through 01:28:07 UTC and failed `15:0` in the first loader probe. Source staging, both runtime identities, invocation capture, and the complete generation-zero input preflight succeeded. Before any training update or rank result was written, all ranks rejected the first batch at `batch_record`: the live Transformers collator returns a `BatchEncoding`, which implements `Mapping` and contains `input_ids`, `attention_mask`, and `labels`, but the diagnostic required the concrete type `dict`. No summary, artifact inventory, completion marker, or scientific result exists. The retained scheduler log and stderr SHA-256 values are respectively

`29939733705f0f3e49dd2fe2e1210484f68a3d6b90f6fafe361fb992cc7718a2`,
`4f5e6fb8ab374b4bce06213cab824ed588b5ee4ae5c6ad818ce4b2f10ee34326`.

The failure is a fail-closed diagnostic type-contract bug, not model evidence.

The corrected execution identity is

commit `f1cfd305cfe76039cf8a0d62881e84ec5faee6d7`,
tree `5c558d3e470b3c5f47aa7912f8ffed71bcbd5255`.

It replaces that concrete-type check with `isinstance(batch,Mapping)`, documents that the retained identity is the two-field training-input projection, and tests an actual three-field `BatchEncoding` plus negative non-mapping/missing-field cases. The focused generation-zero plus replay suite now passes 97 tests. An independent review reproduced the real collator type and found no analogous live-API type blocker. The corrected helper SHA-256 is

`9e9c20f71fb5810ae39da45e29dddaa2c7b130e4b5b7740e819a9a2664cbb63c`.

The runner remains byte-identical. The replacement 4,238-file immutable snapshot is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/`
`rdiag_shared_dt_gen0_repro_diag_`
`f1cfd305cfe76039cf8a0d62881e84ec5faee6d7`,

independently revalidated against source-manifest SHA-256

`524164d0e3b75e91b9cf3013ac581b0d0a12b8288aeec8ec33579efb6eb87a55`.

Replacement two-node H200 job `1835101` ran directly from that snapshot on `h200-081-214` and `h200-199-061` from 2026-09-04 01:57:03 through 02:11:54 UTC and failed `15:0`. Unlike job `1834859`, all three loader modes completed: 48 rank records prove that the two-field training-input projections of all six microbatches match exactly between no-snapshot, snapshot-and-continue, and new-loader sidecar-restore paths on every rank. The L1 and L2 canonical loader-state hashes are also identical. Thus neither calling `state_dict` nor restoring its captured state is the first observed loader fork in this environment.

The first training arm then failed on all ranks before model construction or any update because the shared-DT frontend import loads TileLang, whose TVM initialization intentionally prepends its Python directory to the in-process `PYTHONPATH`. The harness validated the pristine launch environment after that import and therefore rejected the expected library mutation. No training-arm record, summary, artifact inventory, completion marker, or scientific classification exists. The retained artifact root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/gen0_repro_f1cfd305cfe7_1835101/artifacts`.

Its scheduler log, stderr, invocation, and pre-run checkpoint-inventory SHA-256 values are respectively

`1d1770bc94f23f14a4d3d97083827130031eec773131a7512fd4ce2abe350c8f`,
`ccb4628c9a94d73a0f324f748c106c38bc44b846b3e797eef809febb6d3ac9b9`,
`b13029c37f098209ec5eb046e78ecd4652c38af6998fbe9e88fe8caf7be8b45c`,
`20bbc0c2a345eaa46671d670c0d865b6f9a73266c9310bd81a1f43243ed120fd`.

The second correction moves the unchanged launch-environment validation before heavyweight model imports; it does not weaken the check. It is reviewed and pushed as

commit `e2b5e317b750e88437d30e95eb6a89ac88cb11c2`,
tree `f625647a419c9ded1dd02a9b5367827981b00b18`.

The helper and host-test SHA-256 values are now

`fc3c44e57af6d0994934a7bd0c7433580904a481428966773e6863b77adf2217`,
`4be33408b0e33743486be389d31968f3fd6337c83d58f823a9d1ce33860f803f`.

The combined generation-zero/replay host suite passes 98 tests; an independent import-chain audit reproduced the TileLang mutation and found no earlier mutating import. A fresh 4,238-file Git-object snapshot is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/rdiag_shared_dt_gen0_repro_diag_`
`e2b5e317b750e88437d30e95eb6a89ac88cb11c2`,

verified against source-manifest SHA-256

`797858b93cf5b7c1e98581e00bd3fa58b0db28d1de3c87e4b23b3b5a4775b968`.

Replacement job `1835588` ran on `h200-046-019` and `h200-171-121` from 2026-09-04 02:49:35 through 03:07:27 UTC and failed `15:0`. The environment-order correction worked: all three loader phases and the three-update `minimal_a1` arm completed on every rank. Its 16 training records have finite, rank-equal pre-clip norms at all three boundaries and use the same input projection as the restored sidecar. The subsequent cache-inventory step failed before the next arm because the training path, unlike each loader probe, had not explicitly stopped its four persistent DataLoader workers per rank. A local reproduction showed that live workers create `TMPDIR/pymp-*/listener-*` AF_UNIX sockets; the strict cache inventory correctly rejects such a special entry. This is post-measurement harness cleanup evidence, not a model failure, but the job has no final summary, artifact inventory, completion marker, or admissible classification. The retained artifact root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/gen0_repro_e2b5e317b750_1835588/artifacts`.

Its scheduler log, stderr, invocation, and rank-0 `minimal_a1` record SHA-256 values are respectively

`a064b744df142a24957ae3ad3915da441d127d7c37c84c2b6ed91018024f2fe3`,
`f9f48e95fb92bebe27c4f0ddb5ec44f27dc560854f5c28da92ca3c696d812424`,
`356a404e648d383301ac25b79bd6680c9ce289dd49877538fc869bf7a62e9374`,
`b358564f0dfb6f8a39f077258a620119d5030743ce49e4c2c10118a77cf2cf4c`.

The third correction factors the already-used worker shutdown into one helper and invokes it from both loader probes and training-arm cleanup. Special cache entries remain fail-closed, and future errors now include their relative path, decoded kind, and mode. The reviewed, pushed identity is

commit `0e9cda6875e8709fa250110dcb35772d46d22817`,
tree `dad58c03e607bf38304dd41ae7d6e8993d729720`.

The helper and host-test SHA-256 values are respectively a37fdea21c59757246c6fabff0dac15d8ca93ac3cfdc784d0549413544e30f70,

92766a1a21c35be8a26314cf0c4e85b8d298eec88abc37467d6a5e69d632ffa4.

The combined focused suite passes 100 tests, including an AF_UNIX-socket rejection test, and an independent review confirmed that shutdown removes all four reproduced listener sockets without changing measured training math. Its fresh immutable snapshot is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_gen0_repro_diag_
0e9cda6875e8709fa250110dcb35772d46d22817,

verified against source-manifest SHA-256

b228c5e0b44a9888d96d2d381c53940d7f6eb7264c71367ecde06f26535cf0e6.

Replacement job 1835824 ran directly from that snapshot on h200-046-019 and h200-094-010 from 2026-09-04 03:42:39 through 04:13:57 UTC. The job and every step .0–.18 completed 0:0. It produced the exact preregistered closure: 112 rank JSON records across seven 16-rank phases, 214 mode-0400 files recorded by artifact-inventory.json, and 215 final directory entries because the inventory does not list itself. Independent validation returned VALID; the log has exactly one non-authorizing completion marker and the scheduler stderr is empty. The retained artifact root is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/gen0_repro_0e9cda6875e8_1835824/artifacts.

The summary, artifact inventory, invocation, scheduler log, and empty scheduler stderr SHA-256 values are respectively

aa466253b9bcdfeebf695ead220357b6693189c93aa5d089c573a3e887a71df6,

7aef2fc9349c84876b0ea7d46ee114147a4dc52be2cd4b0c776c4d6245b08339,

eba71feda5c6f650968a143f49c4881fa524ce22ac4dec31b713ca1e0bbc9534,

8567a91cdacd6df770a49cc61f13ffe5be0431e479f30bbeae21afb50bdc6f40,

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855.

The result is valid diagnostic evidence but fails the scientific hard gate. All loader and training input matrices match the original sidecars; the reconstructed model, optimizer, loader, first-RNG anchor, rank topology, and corresponding-rank mapping also match. Nevertheless both the minimal A1/A2 pair and instrumented B1/B2 pair classify as ORDINARY_RUNTIME_NONDETERMINISM, making the overall training classification the same. The first classifier-stage difference is the boundary-1 local-gradient Merkle root after the first two microbatches. The earliest chronological scalar difference occurs even earlier at rank 1, microbatch 1: minimal A1 records loss token 0x1.50da340000000p+3, whereas minimal A2 records 0x1.50ddc80000000p+3, despite joint input SHA-256 5dcab61b5c06aa429639e90ef660cba3caf85ed6161fd6cf36812696f369cf7a and pre-forward RNG SHA-256 7db887f05ff2b0af83e6e7c4b4c9d7ac1a71e2f172d903ea409ce54cf21bdabe being identical. Because both minimal replicates diverge before clipping, this is neither a loader restore failure nor instrumentation-only or pre-clip-norm-collective divergence. The exact operator remains unidentified; trigger capture stays blocked pending a focused runtime/kernel repeatability diagnostic and a successful repeat of this hard gate.

That focused forward-determinism diagnostic was implemented in the isolated worktree

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_forward_determinism_20260904

on branch codex/rdiag-shared-dt-forward-determinism-20260904, based on commit

0e9cda6875e8709fa250110dcb35772d46d22817.

Its reviewed executable plan is docs/plans/2026-09-04-rdiag-shared-dt-forward-determinism-plan.md; its entry points are

- scripts/analyses/reproduce_rdiag_shared_dt_forward_determinism.py;
- scripts/analyses/run_rdiag_shared_dt_forward_determinism.sbatch; and
- scripts/tests/rdiag/test_rdiag_shared_dt_forward_determinism_host.py.

The design compares two ordinary-autotune and two diagnostic-only RMSNorm-warp-8 phases. Each process retains one minimally observed cold forward through the normal Fabric/FSDP and health path, stops after the exact ordinary cross-entropy reduction and before backward, then performs three equal-RNG warm traces plus a post-trace replay of all six existing RMSNorm forward configurations. Cold and warm evidence are reported on separate axes; every configuration replay must be stable across corresponding ranks before RMSNorm selection can be called strongly supported. This diagnostic cannot authorize trigger capture: any candidate execution control must still pass a fresh copy of the unchanged generation-zero minimal=MATCHED gate. Independent plan and runtime reviews found and closed three pre-launch defects: an invalid cross-scope FSDP weight-object identity requirement, conflation of the raw BF16 lm_head output with Fabric's distinct FP32 return/loss input, and a missing final fail-together exchange after cleanup and rank-local result validation. The focused suite now passes 57 tests and the combined forward-diagnostic, generation-zero, and soak-790 host suites pass 157 tests. The reviewed and pushed diagnostic identity is

commit 70165b017cd28df832a9a8797a72a1f8234284d0,
tree c58e220299a54525c443f4f0dc204e22e1fb19b8.

The plan, helper, runner, and host-test SHA-256 values are respectively

7aff0bea0a85887e7744f57816239dd598426c5bc4623ffb140a01f410a98f26,

52583f1d8cfdfdd571897a446ce00a69b598a72889e244be9e342498fd32010f,

eeb1704bdb8bb30a376ba5483747e1ad6022426f96fe6cab3e8627cf642fe0fd,

71a20bbcfce4c1109766263bca07e5e8099312f7ec6196d4b2e276aba62a3a65.

The 4,242-file immutable source snapshot is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/rdiag_shared_dt_forward_determinism_70165b017cd28df832a9a8797a72a1f8234284d0,

verified against source-manifest SHA-256

e1129c0caa1832c5bc5ff3d2f978a8c6a18a38a1c55f7d0027b1792ddb77e665.

H200 diagnostic job 1836996 ran from this snapshot on h200-079-003 and h200-079-202 from 2026-09-04 06:17:47 through 06:37:39 UTC. The allocation and every step .0–.12 completed 0:0. It produced the exact preregistered closure: 64 rank JSON records, 64 NCCL logs, eight node/phase cache inventories, 154 mode-0400 files recorded by artifact-inventory.json, and 155 final directory entries because the inventory does not list itself. Independent validation returned VALID; the log has exactly one non-authorizing completion marker and scheduler stderr is empty. The retained artifact root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/forward_determinism_70165b017cd2_1836996/artifacts`.

The summary, artifact inventory, invocation, scheduler log, and empty scheduler stderr SHA-256 values are respectively

`741e75362c6214b4a82644bbab2b12b61fc6ed827e80ff4ac4d3f86be94861ff`,
`463b68fc3d70f85682fe03137df857e5e577d9c9aae009e7f7c13659f591931f`,
`0179440c6f58ac1935626684916b4599bd2035820cde6ee4ccf1cc5ebcdbfde5`,
`45a7018391bd5ec572d48c2557d3a0239cadd679289cf465f8e241a444cfb29a`,
`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`.

The pre/post generation-zero input inventories are byte-identical, both with SHA-256 `b5bc763929a966c54adbfeb1d9a1d9d1c0ae891556860ad89449a333fec50cbe`.

The preregistered classification is `RMSNORM_AUTOTUNE_SELECTION_STRONGLY_SUPPORTED`. Every initial-state and input comparability axis matches, every within-process warm repeat matches its cold result, and both warp-8 phases match exactly across processes. By contrast, the two ordinary cold processes diverge on ranks 3, 6, 12, and 14. Those are exactly the ranks whose selected forward RMSNorm warp count changes: respectively $8 \to 4$, $2 \to 4$, $4 \to 8$, and $4 \to 1$. Their first warm-trace difference is `block_00.norm_1`; their external logits, unreduced loss, and reduced loss then differ, while z-loss remains equal. Replaying each selected configuration reproduces its recorded warm output, and all six forced configurations are stable when compared at equal input, weight, rank, and configuration. Thus timing-dependent RMSNorm forward autotune selection is strongly supported as the source of the observed first-forward variance. This remains diagnostic rather than production authorization: it does not test backward repeatability, and a narrowly scoped fixed-warp production control must pass a fresh unchanged generation-zero hard gate.

The corrective implementation is now isolated in worktree

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_worktrees/rdiag_shared_dt_rmsnorm_control_20260904`

on branch `codex/rdiag-shared-dt-rmsnorm-control-20260904`, based exactly on diagnostic commit `70165b017cd28df832a9a8797a72a1f8234284d0`. Its executable plan is `docs/plans/2026-09-04-rdiag-shared-dt-rmsnorm-determinism-plan.md`. The selected scope is the narrow forward-only control justified by job `1836996`: for the four resolved shared-DT configurations only, require the cold RMSNorm forward autotuner's exact existing six-candidate closure and install its unique existing warp-8 object as a singleton. The backward RMSNorm tuner, every FLA tuner, precision policy, recurrence, parameters, and generation-zero helper/runner remain unchanged. Activation keys on the resolved internal config identity rather than the raw CLI name, because the generation-zero diagnostic deliberately uses a temporary CLI alias whose copied payload retains the production internal name. No environment selector, config/recipe field, or runtime-fingerprint field is added: the historical runtime, data, model, and sidecar identities remain fixed, while the new source-manifest SHA binds the control. The implementation must fail closed on warm, altered, or ambiguous tuner state, emit a canonical policy digest with rank consensus before model construction, cover the direct evaluation and gradient-gate constructors that bypass `pretrain.py`, and prove that the backward tuner is not mutated. The process-global pin is intentionally irreversible; every authenticated entry point runs one shared-DT model family per process, and a mixed-family caller must use a separate process rather than attempting to restore Triton global state.

The initial policy implementation and smoke harness were reviewed, committed, and pushed as

commit 28ea1c51a16825ae02d3d4c6520087e8b9b2ca42,
tree 1f7dd60edbebe92b429e4a541fdea85e6ec3c182.

The plan, policy module, trainer integration, authenticated evaluation integration, and gradient gate integration SHA-256 values are respectively

91cc8fd7580ff953f71cedd5594a489eacc126168bce68904d7b17ffebbcbbb2,
74898f2dfa3c3073e491616eb7e847acdaa695c4b8b70596af5534e5c6591433,
ff9fa812d81266577c5d9b4404966e039e1bad205abf19009099ffc24651a584,
b78e03c7996933e5c68a2e08a46ab074a8a235b5ac783e2588a523e14497c5f8,
755fa87cc6598b21391a06de20a9b25ce1a9286b383e8e714a26764c141015e6.

The corresponding implementation entry points are

- lit_gpt/rdiag_shared_dt_rmsnorm.py for applicability checks, fail-closed tuner inspection, installation, and the canonical policy record;
- pretrain.py, scripts/eval/eval_lm_harness.py, and scripts/tests/rdiag/rdiag_shared_dt_gradient_ab_run.py for authenticated training, evaluation, and gradient-gate activation; and
- scripts/tests/rdiag/test_rdiag_shared_dt_rmsnorm_policy_host.py, scripts/tests/rdiag/test_rdiag_shared_dt_rmsnorm_integration_host.py, and the policy smoke files below for host and H200 contracts.

The smoke helper and runner are scripts/tests/kernels/test_rdiag_shared_dt_rmsnorm_policy.py and scripts/tests/kernels/run_rdiag_shared_dt_rmsnorm_policy.sbatch, with SHA-256 values aa58e4cfa0efb2f35e4cf4c38d86eb1492a1fbba359b644c7cf526c8c15e13cc and 85f8ce9bf11cf61d006916ac44c3955fbe1f7b1c070ac17453f4afd89f850c84. It exercises the actual wrapped Triton forward at $M = 16384$ and widths 1024, 2048, and 2304, requires repeat and two-rank output/rstd byte equality, and benchmarks all six existing candidates with 40 order-balanced samples each. Warp 8 must remain within 1.25× the empirical fastest candidate by both median and lower-quartile latency. Installation leaves the backward candidate objects unchanged; a later backward smoke only proves that the untouched path still executes. At the initial implementation revision, the focused policy/integration/smoke contract suite passed 49 tests. The combined existing generation-zero, forward-diagnostic, runtime-identity, gate-runner, and authorization suite passed 309 tests with three declared skips when pytest temporary files were redirected to Lustre; an initial attempt exhausted the already-full system temporary filesystem and is not a code failure. After the later smoke-helper and launcher-only revisions, the final focused runner/smoke-host suite passed nine tests. Independent final review reports no Critical, Important, or Minor finding. The generation-zero helper and runner remain byte-identical to commit 70165b0, with SHA-256 values a37fdea21c59757246c6fabff0dac15d8ca93ac3cfdc784d0549413544e30f70 and f9d5c8a04f7384b360eb04435c32d0a2abeb177e6f6534e91f4910e5b7c1bb14.

The 4,249-file immutable control snapshot is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/source_snapshots/rdiag_shared_dt_rmsnorm_control_28ea1c51a16825ae02d3d4c6520087e8b9b2ca42,

independently verified against source-manifest SHA-256

4207beb5e9eb0aeaaa11c7eb489d2295122ac257ecedfc77b5dfde21f35241c1.

Non-authorizing two-H200 smoke job `1837294` was submitted from exactly this snapshot and failed before Python after two seconds. The batch shell and its `srun` task steps receive different node-local `/tmp` namespaces on this cluster; the batch-created cache parent was therefore absent in both GPU tasks. No numerical test executed and no result JSON was published; the empty artifact directory is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/rmsnorm_policy_28ea1c51_ZxLwOVLa/artifacts`.

The runner correction creates and removes a distinct private `/tmp` root inside each `srun` task, preserves the workload exit status through guarded cleanup, and has passed its seven host contracts, shell syntax check, and independent review. Because this changes the runner bytes, the reviewed and pushed replacement identity is

commit `edf5bb3aa95e2d64beabc8f4196c56029851b037`,
tree `fbcc46493836860a268d9fe56514ca8eadd38993`.

The corrected plan, runner, and runner-host-test SHA-256 values are respectively

`cb77a67a4e06e5d31c93547e4560064eb10bd045261a37a8bc5b971e7338fdb7`,
`48463cff46aa3eae19a46514655dfac90b8886e02a8df08036a9df4ad12c66ba`,
`2b06e25a26341d7eb108f06c6abe7c087a7ea035d9ede29715eb36f801814b75`.

Its independently verified 4,249-file snapshot is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/rdiag_shared_dt_rmsnorm_control_`
`edf5bb3aa95e2d64beabc8f4196c56029851b037`,

with source-manifest SHA-256

`2dc7fc4d3856c43f607e00df606a63c4c221fcbd5c5da5f461e6fab34fc59c98`.

Replacement smoke job `1837353` reached Python from this corrected snapshot but failed on the first NCCL collective: task-local one-GPU visibility caused NCCL 2.27 to report `invaliddeviceordinal`. The irrecoverable allocation was cancelled after 2:08 to release its H200s. It also published no result; its empty artifact target is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/rmsnorm_policy_edf5bb3a_MBXIibLe/artifacts`.

The second launcher correction follows a completed project H200 reference: both allocated GPUs are visible to both tasks via `--gpus-per-node=2--gpu-bind=none`, and ranks select CUDA devices by `LOCAL_RANK`. It also prints rank-local phase tracebacks before recovery and re-raises fatal CUDA/NCCL failures instead of attempting another collective on a poisoned CUDA context. The reviewed and pushed identity is now

commit `fd9ca79e697b2fce14b8938459a7d2e63e434bea`,
tree `b72796b9106c8e06d2b7e59051e5cab094ec8b40`.

The current plan, smoke helper, runner, and runner-host-test SHA-256 values are respectively

`61277a3f3db22a8495f5cc2cfbdd50fd038cc53664c219c04bcacad8e8f3c572`,
`af2f17b7b178930291a1225345373713c3fc1357165a3812afa9f237c494d31e`,
`87652ee3fdb1fd54cf17cfdbcb32e65877aa1fa27b8c146af96b860e0fe9d38f`,
`cfb736f894504e8bc2f941074452579dbb87c825ce5fd7cb54c36e8674821d69`.

Its independently verified 4,249-file snapshot is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_rmsnorm_control_
fd9ca79e697b2fce14b8938459a7d2e63e434bea,

with source-manifest SHA-256

0d49e9471d3e6a3b48bc21838b6366cf26d7b8f9418639b70a5f012dec866f20.

Smoke job 1837393 ran from this snapshot on h200-129-161 from 2026-09-04 07:51:39 through 07:52:08 UTC. The allocation, batch, extern, and sole srun step all completed 0:0. It published exactly one retained artifact,

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/rmsnorm_policy_fd9ca79e_gz8Nu1Ly/artifacts/result.json.

That mode-0400, single-link, root-owned result is 17,835 bytes and has SHA-256

f39137e97c07d7cafc9d930b4060dce798b352a439aba9a22201d106107aeb30.

The scheduler log SHA-256 is

a6228b2518a21a27e627c69c2f0b61d96613396230833eaa7acd2c757b288490.

Independent validation returned VALID. The result is explicitly authorizing=false and production_gate=false. Both ranks agreed on policy record SHA-256

cd31a0f1bc60aa5421cb7074f5236a14ff4fb66d59632e587e0920f4e343ecac.

At widths 1024, 2048, and 2304, the first forward used the exact pre-existing warp-8 object, the forward cache remained empty, repeat outputs and reciprocal standard deviations were byte-exact, and the corresponding cross-rank values were byte-exact. Across all rank/width pairs, the maximum warp-8-to-fastest ratio was 1.04614 by median and 1.04703 by lower-quartile latency, well within the preregistered 1.25× bounds.

The deliberately untouched backward smoke also exposed the next concrete risk. Rank 0 selected warp 16 while rank 1 selected warp 8; equal forward outputs and weight-gradient hashes accompanied different input-gradient hashes:

f364b0deeac41662c4571b76f4ae57c9d45ac5923b47b0d0685f357ca7f83647,
ae9fca3f6523a4a59d0670dd4af9bdbf041c6992f5df43eb37907baa1a9be3d3.

Backward cross-rank equality was not a preregistered requirement of this forward-control smoke, so the bounded smoke validly passes; the observation nevertheless makes RMSNorm backward autotuning the leading suspect if the forward-only control is insufficient. The unchanged replacement generation-zero run discriminates that sufficiency, not backward causality. That gate, not installation or this smoke, decides whether trigger capture can begin. If it does not return minimal=MATCHED, focused backward-RMSNorm localization is the next priority, and a separately reviewed narrow control is warranted only if that localization implicates it. No trigger or production authorization is presently claimed.

The unchanged replacement generation-zero job 1837446 then ran from the same fd9ca79e... snapshot on h200-004-175 and h200-071-008 from 2026-09-04 07:57:40 through 08:27:18 UTC. The allocation, batch, extern, and all steps .0–.18 completed 0:0. Its retained artifact root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/gen0_repro_fd9ca79e697b_1837446/artifacts`.

Independent validation returned `VALID`: the directory contains the exact 214-file inventoried closure plus `artifact-inventory.json`, for 215 mode-0400, single-link files. The summary, inventory, invocation, scheduler log, and empty scheduler stderr SHA-256 values are respectively

`c69adb84b265f7c20560ad1be54a7654fcda963ffe068cda252f398c942d9c54`,
`97a46cae0bc31162351ffa1d39acf6f55b048d4f8ccc22677b549082a3c64a14`,
`2bf614563fa4f82662220809cfaf2c1b9d23f512e11400e87fa11db40644f179`,
`fb0cbcfc30f950f5b899095bb603caf1110cb369eb25eff6bdba682d396b1e90`,
`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`.

The log has exactly one completion marker. Its pre/post checkpoint-input inventories are byte-identical, both with SHA-256

`20bbc0c2a345eaa46671d670c0d865b6f9a73266c9310bd81a1f43243ed120fd`.

Both node runtime-identity records retain SHA-256

`135ea68e47d1e9728f1295e480dd22518566f818619d06b6853bed343b00639d`.

The hard gate passes. The reconstructed common anchor, rank topology, training inputs, and sidecar inputs match; all three loader arms match over their first six rank-local microbatches. Both preregistered replicate-pair classifications are `MATCHED`, including the required `minimal=MATCHED`; global loss and pre-clip norm are exact at each of the first three update boundaries. The overall training classification is `MATCHED_RECONSTRUCTED_FIRST_THREE_UPDATES`. This remains non-authorizing and does not make the reconstructed process RNG historically comparable to the failed production run, whose RNG was not retained. It does establish that the forward-only control is sufficient for the generation-zero repeatability gate. The backward-smoke observation remains a risk note rather than evidence warranting a second intervention, so no backward pin is introduced. This result released revision, review, and non-authorizing execution of the trigger-capture diagnostic; it does not authorize production.

The gated follow-on capture work is isolated in worktree

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_worktrees/rdiag_shared_dt_trigger_capture_20260904`

on branch `codex/rdiag-shared-dt-trigger-capture-20260904`; its plan is `docs/plans/2026-09-04-rdiag-shared-dt-trigger-capture-plan.md`. Its earlier plan-only identity was

commit `8074daf35bbd3dac839f0fc2601361d7cf069d50`,
tree `b27a3f6ab3269df09daf02f08507f178a77cefe9`.

The accepted control lineage was merged without rewriting either parent as

commit `57afdd1f9fe497f3630f262f7febfcbbd56e4f7e`,
tree `ad12554140dac65bb315dd49a0c7fc003393acb1`,

whose parents are `8074daf...` and `fd9ca79e...`. The plan was then revised, independently reviewed for evidence integrity and runtime semantics, committed, and pushed as

commit `dfd928cf7e861f5f828f6486730cf2836cdd3089`,
tree `1666a25b3ef188dca6c151a18209c11d57b470f9`.

Its reviewed plan SHA-256 is

48f65aaf84d0c0ce403ed25bc9bd860c33b511c2119c28191b3e71123473e7d8.

The plan pins job 1837446's summary, inventory, invocation, source, input, runtime, and log identities at their actual nested classifier paths. Its exact 1,281-byte root scheduler row is retained separately, without modifying the closed generation-zero directory, at

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/gen0_repro_fd9ca79e697b_1837446_scheduler_record/
sacct-root-row.txt,

with mode 0400, one link, and SHA-256

7b3df5231f751e1841c448cb55714c471522bcd74bbb973b577477785f07f587.

Review also closes process-RNG changes during loader read-ahead, rank-asymmetric capture failures, unexpected loader EOF/worker errors, nonfinite-forward handling, live RMSNorm autotuner provenance, and the two earlier launcher topology/cache failures. The executable scope remains the plan plus scripts/analyses/capture_rdiag_shared_dt_trigger.py, scripts/analyses/run_rdiag_shared_dt_trigger_capture.sbatch, and scripts/tests/rdiag/test_rdiag_shared_dt_trigger_capture_host.py. The implementation was independently reviewed, committed, and pushed as

commit ef57031fc2f342a0dfe9c9ef7af2d0e268e332cb,
tree db686e18708233ecbb049cb68cbc1ace9d81d4d6.

The initial helper, runner, and host-test SHA-256 values were respectively

ec0db20eb6f0609425c5add313dab82c111b7619e8c2c0f94840ed45cc0e29d3,
11c92b07965f69e622fe7ec0af3ae00256942eaeb7c9192adda7d15cf6cde95e,
56eb9d0a04d053a0e1e19724155e56dc3f1d225ee150c4d18bbbf8c19127f8a8.

The initial focused suite passed 60 tests, the adjacent RDiag suite passed 170 tests, and the broad compatibility sweep passed 495 tests with three skips. Its immutable 4,253-file Git-object snapshot, excluding only the paper gitlink, is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_trigger_capture_
ef57031fc2f342a0dfe9c9ef7af2d0e268e332cb,

with source-manifest SHA-256

5d3edaa28885ee012fde8887f96af827f0fe320b84eba4ea5e0b5ff46cccb907.

Both the publisher's closed-tree verification and a fresh private-copy reconstruction from Git objects passed. The filesystem's unavailable no-replace rename path required the publisher's fail-closed copy fallback; the final execution root is owner-held mode 0555. Hidden staging siblings retained by publication and by a concurrent collision check are evidence only and are explicitly rejected as execution roots.

Non-authorizing two-node, 16-H200 trigger-capture job 1838085 was submitted from that snapshot at 2026-09-04 10:11:12 UTC with the exact failed-run data manifest. It ran on h200-205-088 and h200-231-137 from 10:30:02 through 10:33:51 UTC, then failed closed before capture training. Source-copy, topology, runtime-identity, and two-node exit-cleanup steps all completed 0:0; the root and batch exited 1:0. Its run root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/trigger_capture_ef57031fc2f3_1838085`,

where `anchor` and `output` are empty and `artifacts` contains only six preflight files. There are no capture results, NCCL logs, summary, inventory, checkpoint, or completion marker. The external root and full-step scheduler rows have SHA-256 values

`2d9eae9ab2fc00dd7e407629b9214d97efb4629fb1753f5bbac9b5f0ebf9380f`,
`395f5ebda2a1ad3910b956209ead563a3de974fd80f9a72a1a27b6fec76105db`,

under `diagnostic_runs/trigger_capture_ef57031fc2f3_1838085_scheduler_record`. Scheduler record filenames are `root-row.txt` and `all-steps.txt`. Scheduler stdout and stderr are

`/lustre-storage/checkpoints/ngocbh/hdn/ckpts/logs/`
`rdiag_sdt_trigger_1838085.log`, and
`/lustre-storage/checkpoints/ngocbh/hdn/ckpts/logs/`
`rdiag_sdt_trigger_1838085.err`;

their SHA-256 values are respectively

`158b30dd9c16f898aed9b40df8272e059506c806384c8a75359ba919fe5a5447`,
`196a496ecaa4e47cba69b79bab5292c81d2006f65de9861d43c47701623cba87`.

The defect was a strict framing mismatch rather than a scientific failure: the helper queried `sacct-P` (`--parsable2`), whose 19 fields have 18 separators and no trailing `|`, but parsed it as `-p` output with a trailing separator. The retained 1,281-byte job-1837446 row and a fresh `-P` query remain byte-identical with SHA-256

`7b3df5231f751e1841c448cb55714c471522bcd74bbb973b577477785f07f587`.

The minimal correction keeps `-P`, requires exactly 19 fields, and uses a strict zip; it does not alter any scientific path. It was reviewed, committed, and pushed as

commit `4cbddbc1f715e5eb3a4177fe1557b29d7ab4f3d0`,
tree `869a5f036fe8cf2bcb085c8b4e8d38cb5e64f550`.

The corrected helper, unchanged runner, and corrected host-test SHA-256 values are respectively

`3d6879298bed98dfc6fa992d518d7d0ef0ae80267c4e8d5c7db3a134ba1abca6`,
`11c92b07965f69e622fe7ec0af3ae00256942eaeb7c9192adda7d15cf6cde95e`,
`8eacb854d8d3297c2e9f1eeb2a19a404f51521f900027e29af47f3ff3957602d`.

The corrected focused suite passed 68 tests and the broad suite passed 503 tests with three skips; the full real prerequisite command also passed against the pinned checkpoint, generation-zero closure, data manifest, scheduler record, and initial snapshot. The replacement immutable snapshot is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`source_snapshots/rdiag_shared_dt_trigger_capture_`
`4cbddbc1f715e5eb3a4177fe1557b29d7ab4f3d0`,

with source-manifest SHA-256

`e27fb7c3a64465ddd3b4d92cbd9d3dc0e83d0e2f9d5a48e8b69540e7f9327b6d`.

Both publication verification and a fresh private-copy reconstruction passed. Replacement job `1838250` was submitted at 2026-09-04 10:48:12 UTC, started at 12:21:08 UTC on `h200-094-010` and `h200-094-088`, and failed closed at 12:41:26 UTC. The two-node source-copy, topology, and runtime-identity steps completed `0:0`; exact snapshot, generation-zero closure, data-manifest, and retained scheduler-evidence verification all passed, so the corrected parsable2 boundary has been exercised successfully. The intended 16-rank capture step started at 12:24:44 UTC, reconstructed the 621,284,352-parameter model, resumed the exact generation-zero state, and trained normally with 60.4 GB peak allocation per GPU. Update 835 was the last printed completed update. An eligible finite, rank-agreed trigger with norm in $[3, 10]$ subsequently became pending before update 840, but its exact boundary and norm existed only in process memory and are not recoverable from this failed attempt. During anchor publication, all 16 ranks then failed together in `anchor-rmsnorm-tuners` before any anchor write: `inspect_rmsnorm_tuners` passed the fully qualified display strings `lit_gpt.rmsnorm._layer_norm_fwd_1pass_kernel` and `lit_gpt.rmsnorm._layer_norm_bwd_kernel` directly to module `getattr`, although the module exposes their basename attributes. Capture step `.3` therefore exited `1:0` after 16:39; cache harvest/removal step `.4` and private-source cleanup step `.5` completed `0:0`. This is an implementation failure after trigger selection, not `NO_TRIGGER_THROUGH_900` and not a scientific trigger result. Its deterministic run root is

`/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/`
`diagnostic_runs/trigger_capture_4cbddbc1f715_1838250`.

It contains 43 mode-0400 partial artifacts, an empty anchor root, and only the expected zero-byte checkpoint placeholder under `output`. Rank capture results, `capture-phase`, post-input inventory, invocation, summary, artifact inventory, loader verification, and completion marker are all absent. The scheduler stdout/stderr and capture stdout/stderr SHA-256 values are respectively

`2fae5175174f72c705af06fb0023a333785f447f9d83214f00e126d921a4aa1b`,
`028a4dacfc45125ba976c3990402c20a538e306929c352f6b8178d99351f5d8a`,
`1d39eebde75592b3a4fa202a37b0be5f122884f05cbfcba34b9a50b8c42d24c3`,
`aa7dbceb04a817036fafc5192c1b32a7e921357e7348ce7ac2fce3c8f4567e38`.

The externally retained root and nine-row full-step scheduler records are mode 0400, one-link files under

`diagnostic_runs/trigger_capture_4cbddbc1f715_1838250_scheduler_record`,

with SHA-256 values

`3db33e5c49a94e7fcb7dfef72a19f9b4d6c67d0597a2108940e140eaf24a870e`,
`012202c4d97ce0ae5a7ccbcccc002e2bef202ca192cb182f8f0ea51bd1b96185`.

The minimal repair resolves each qualified name's final component for Python lookup while retaining the qualified string in canonical policy evidence. A host regression now exercises the real qualified-symbol/module-basename shape and proves inspection leaves both tuner states unchanged; the focused suite passes 69 tests and the focused-plus-policy suite passes 91. Independent review found no blocking issue; Python compilation, shell syntax, and Git diff checks pass. A deliberately overbroad login-node sweep was not an acceptance test: its CUDA cases rejected the GPU-less host and one unrelated source-publisher test observed the already documented Lustre staging fallback. The reviewed and pushed repair is

commit 015ba66e07f0ec1860b188ea74e638ce955384cc,
tree d1d59c7a08112e29ac942b0153f36552a7cc60ea.

The repaired helper, unchanged runner, and regression-test SHA-256 values are respectively

c732721ae49d4798a2ff280300ffb09f7e4f83470ee415092e8955dd7f51ca5e,
11c92b07965f69e622fe7ec0af3ae00256942eaeb7c9192adda7d15cf6cde95e,
02ec5161fbb73dbe1db0a1ab197fc9cf908844fa04c1871a48ac8e9e1ad726ef.

Its immutable 4,253-file snapshot, again excluding only the paper gitlink, is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_snapshots/rdiag_shared_dt_trigger_capture_
015ba66e07f0ec1860b188ea74e638ce955384cc,

with source-manifest SHA-256

bf0b5288a873f5f7c67963492239226921150b2c0f7350de6a3cf57367987dca.

Both publication verification and an independent fresh private-copy reconstruction passed. Replacement job 1838750 was submitted from that snapshot at 2026-09-04 13:06:42 UTC and ran on h200-094-010 and h200-094-088 from 13:09:11 through 13:45:29 UTC. The root, batch, extern, and numbered steps .0–.8 all completed 0:0; the numbered task counts were exactly $2, 2, 2, 16, 2, 16, 2, 2, 2$. Its sole completion marker is present and scheduler stderr is empty. The preregistered first eligible trigger was absolute boundary 838, with rank-agreed norm 3.4058520793914795, canonical token finite:0x1.b3f2f60000000p+1, and zero health failures. It retained the post-update generation

step-000000000838-iter-000000001676-6ee7296afb63448f9863efab3454d49d

plus 16 RNG records and 16 future raw microbatches per rank. All 16 independent loader-replay records match every ordered joint and component-tensor hash. The continuation later stopped non-authorizingly at boundary 853 with finite norm 11.937856674194336; that postselection stop does not invalidate the retained boundary-838 anchor.

The accepted run root is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/trigger_capture_015ba66e07f0_1838750.

Its summary, artifact inventory, invocation, byte-identical pre/post input inventory, capture phase, derived anchor manifest, and checkpoint-generation manifest SHA-256 values are respectively

0123887d03356ca08a0e88aa267746ac1d4680b4c17a6fb63f98800175bb9b33,
0709972f30d9b4118e41865d8ecc03a55dda9f34bdc856d87d3ef23cbee48500,
04f0d067c7c52cac583f9a5ca528282a5c757c560e6135a8007eedb093586509,
729aee6bea81535c49a5c16092750d0ff603275b866fdfca662abf7ca23d3231,
05b5070b22064b56e1f3aba9403099824c45c338d3c5c6b676cb5918a89461f2,
1c01aa9798bcf3c1d75cc016bd103cac919ca7c8296cddffb4817bede8353bd8,
3b41e9f4590909f461b89aaec0041ff8b7f51347ae531bcd07faf18dba36d008.

The artifact directory has exactly 164 inventoried files and 165 physical files including the inventory, all mode 0400 and single-link. Its decision is CAPTURE_RETAINED_NONAUTHORIZING with CAPTURED_LOADER_REPLAY_MATCHED. The external scheduler-record root is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/trigger_capture_015ba66e07f0_1838750_scheduler_record,

whose root-row, full-step, completion-marker, and scheduler-log-hash records have SHA-256 values

6987ed8bf54cc5b80713503dce990fad621032a4c0d2a4d3c1116c1fd0828494,
3afc238b32b6631cba1bd64b15dd29f012c366c2f47b60789ab754fa57e8428a,
f00be3ff84a1349b7bb29cb5a5460de217f3c932d7389d736e79809dc6d6b6ba,
7fb33026b6196a9d9321774c1650e1d41f302c4ff60a8dcc1e3f5a9b2fc25773.

Scheduler stdout is fb69191ebd7bbc42e9a20222629ae4f03525badc5df4544fb59871f02c5b2b50; stderr has the empty-file digest.

A separate read-only validator, job 1838966, then ran on h200-002-213 from 13:52:43 through 14:01:15 UTC and completed 0:0. It reconstructed the invocation-bound historical path /tmp/rdiag_sdt_trigger_1838750/source, verified the immutable 4,253-file source before and after validation, revalidated source, generation zero, and anchor, returned VALID with artifact_count=164, removed both owned node-local roots, emitted one marker, and left empty stderr. Its evidence root is

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
diagnostic_runs/trigger_capture_015ba66e07f0_1838750_independent_
validation.

The retained validator runner, validation result, acceptance result, root-row, full-step, completion-marker, scheduler-log-hash, and complete evidence-list SHA-256 values are respectively

ce3ee70f2d5f80a9b98ab7fc692ad93d139d3d319115e4b3b889ff8478f2b600,
2f50dd577f9b5a507280c9ec913608b22f8f56a7028dc0f852a66848b544be42,
7e564d9f4131bbc820390cc798929739d8e291cb9535d1196e3a6e64d1511314,
254091248607cbbd8bb476ae5350029974fdb6be1d287fe889cf91c7dece84c9,
acde647a47074436c368fcde4cce661b49ae4bb250a53786d87a0f00e73a0444,
0a2257b4e2b4f8c063059b52c7be11d353a2fa39a547c448d253bf6c4f247565,
d455f9590ade96882b0cffe52132ef96977202c5e26bd559dbf5a084907647d3,
35e2822bc0a296487522dc3f6fe8a10b17cbe196085c5227b2e095425c393cbb.

This closes the trigger-capture prerequisite for implementation; it remains diagnostic and cannot authorize production by itself.

The downstream matched-intervention design and implementation are isolated in worktree

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_matched_intervention_20260904

on branch codex/rdiag-shared-dt-matched-intervention-20260904; its plan is docs/plans/2026-09-04-rdiag-shared-dt-matched-intervention-plan.md. The original reviewed and pushed plan-only identity was

commit 2b8066717dc05128a37404def1dcfa643fd0ef7d,
tree 6aed24664104fa61833f95a7b5cc3626f1baea5d.

It requires one immutable post-trigger anchor and sequential fresh shared-A1, connected-zero no-rotation, and shared-A2 arms over 16 retained microbatches per rank. Exact A1/A2 equality is a precondition for interpreting the intervention. Review corrected the initial draft to compare rank-local FSDP state only with the corresponding rank across arms, use replica-aware global gradient diagnostics, limit first-boundary DT/alpha equality to the treated layer, stop only after the eighth update is fully committed, capture nonfinite-loss exits artifact-locally, retain the ordered 32-loss boundary estimand, and bind all 256 rank/batch joint hashes. That original plan's SHA-256 is

`17a7d50a0349ed1038b4c703dfe25a12e791e5da40c194b5493a46ee087ca0db.`

At that stage, a read-only merge preflight found no textual conflicts, and the branch deliberately did not absorb the plan-only trigger tip; it remained gated on a successful final trigger capture and repin before implementation.

Before observing the trigger outcome, the plan was tightened to pin accepted generation-zero job `1837446`, current trigger commit `4cbddbc...` and manifest `e27fb7c3...`, and the still-pending retry without claiming its result. It now fixes the 32-loss ordering and arithmetic, phase-dependent unavailable fields, value-exact zero semantics, first-boundary frontend scope, early-stop estimands, operational-failure continuation, nonfinite loss interception, and reconstruction of the trigger invocation's historical source path. The reviewed and pushed revision is

commit `444946e6b1f588c4bcfffc083c54ac80ac1cca6d`,
tree `ffa896e55eebdb48f15c4b71ae2bb89da117c2e3`,

with plan SHA-256

`43dc150759e293f39e810f42cc4d6dae11909d5ce28d10f4753a5fa6c2eefa51.`

The acceptance clause explicitly requires both a structurally valid trigger closure and `CAPTURE_RETAINED_NONAUTHORIZING` with `CAPTURED_LOADER_REPLAY_MATCHED`; a valid no-anchor or loader-mismatch outcome cannot release implementation. Because revision `444946e...` pins failed job `1838250`, it is now a historical blocked design record. The reviewed plan was repinned without implementation to pending job `1838750`, trigger commit `015ba66e...`, tree `d1d59c7a...`, manifest `bf0b5288...`, the exact historical source path `/tmp/rdiag_sdt_trigger_1838750/source`, and the corrected retained-anchor closure of 164 inventoried files (165 physical files including the inventory). That reviewed and pushed plan is

commit `41de0ee4836fa492e560f0d1640f6cac73c13382`,
tree `4ebdfc9335b9f4c2f11c6e2c86e98a48eca21587`,

with plan SHA-256

`e7a4df5999f17cae172d9de2f454299fca38879773e225690a14c2aed2c91ff8.`

That repin was the last pre-outcome design state. After both job `1838750` and external validator `1838966` passed, the accepted trigger source was merged without conflict as

commit `9d65a587be64f0a7dc9b2c3b3ed40ac0351c08ae`,
tree `53136608092429aad484efcaffed22cb021233cf`.

The plan was then independently checked against every retained job, source, scheduler, artifact, anchor, and validator identity and pushed as

commit `285c36b9e75c897521e6529cd0e5766c2fcafbb7`,
tree `3787abb2b4480cf73cc581e718a3558706a6dd5d`,

with plan SHA-256

    3b869d76eb17fafcca9ddb1364524927be4a924642f726d8a27649b076c98355.

The reviewed implementation is commit

    49f06078d19f349ca9724e3bbfc08fa436e91e54,

with tree `ebb54a1719a6bf8490e9d6997ec471b5201423f3`. Its implementation paths are `scripts/analyses/rdiag_shared_dt_matched_intervention.py`, `scripts/analyses/run_rdiag_shared_dt_matched_intervention.sbatch`, and `scripts/tests/rdiag/test_rdiag_shared_dt_matched_intervention_host.py`. The final plan, helper, runner, and host-test SHA-256 values are respectively

    13d5bf40a3c5f5974a64a91b4cf3ae72efd999e1bbf012f2e57fc3bac32643c3,
    0a339701716fcb5d404deecd849d36a3cbd40eee654520a229801de90204b180,
    ceaae69aacba7b12325188c7b50d2e503977968f0022b777cfc79c331b40f935,
    09002a176e6eceadf504746b774d0dcf50db08b02b616cc53c5e40c722964482.

All implementation steps before publication are complete. The focused host suite passed 126 tests, and the matched plus trigger, generation-zero, and soak-replay sweep passed 295 tests; Python compilation, shell syntax, diff hygiene, direct validation of all 16 retained rank-local initial states, and live scheduler revalidation of jobs `1838750` and `1838966` also passed. Independent scientific, distributed-runtime, and immutable-artifact reviews found no remaining code blocker after repairs to collective forwarding, post-update evidence, rank-local failure coordination, exact anchor binding, nonfinite-loss rederivation, contiguous-prefix estimands, and fixed-order arm continuation.

The immutable source snapshot is

    /lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
    source_snapshots/
    rdiag_shared_dt_matched_intervention_
    49f06078d19f349ca9724e3bbfc08fa436e91e54.

Its `SOURCE_MANIFEST.json` SHA-256 is

    45644d7de85703ceacfbd4789925245097b00332f1c9a9c5df8da334ce8ce3a5.

The manifest contains 4,257 regular Git files; the physical frozen tree contains 4,258 files including the manifest and 409 directories, with directories and executables mode 0555 and other files mode 0444. The `paper` gitlink is recorded and intentionally excluded. The publisher's built-in verification and a separate reconstruction into a private source tree both matched the commit and manifest exactly. Because Lustre used the supported no-replace fallback, the complete read-only hidden staging tree is retained as publication evidence; it is not an execution source.

Exactly one two-node/16-H200 matched-intervention allocation, job `1839531`, was submitted from that named snapshot at 2026-09-04 15:37 UTC. It ran on `h200-071-128` and `h200-074-040` from 15:59:07 through 17:06:49 UTC and completed `0:0`. The root, batch, extern, and numbered steps `.0`–`.16` all completed `0:0`; the numbered task counts were exactly $2, 2, 2, 2, 2, 16, 2, 2, 2, 16, 2, 2, 2, 16, 2, 2, 2$. The run executed fresh shared-A1, connected-zero no-rotation, and shared-A2 arms in that order, with no requeue and no production authority. Its run root is

    /lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
    diagnostic_runs/matched_intervention_49f06078d19f_1839531.

The in-job final validator returned `VALID`, revalidating the matched source, accepted trigger anchor, and all prerequisites. The artifact inventory contains 118 closed files and the directory contains 119 physical files including the inventory; every artifact is a mode-0400, single-link regular file. The summary and inventory SHA-256 values are respectively

```
140ca76485e6838c6ae3f6bfc086e0112aae5c48f809cfaa06adc92724efe83c,
bdd78e22df0ac67242c36f559b9ede1c1ae3485b575bc3647d25285599d62765.
```

The validity checks passed exactly. A1 and A2 are byte-identical under the complete common-arm projection, with `a1_a2_first_difference=null`; initial state, topology, all 256 rank/microbatch inputs, and pre-forward RNG records match. All three arms committed eight updates through boundary 846. The no-rotation arm has exact-zero angle projection output, phase increment, phase, and current angle gradient while the radial gradients remain live; layer-0 `DT` and `alpha` match the normal arm exactly at boundary 839. Consequently the summary reports `intervention_isolation=MATCHED`, no isolation failures, and causal estimands on every boundary 839–846.

Table 2: Matched boundary-838 intervention, job 1839531. "On" is the exactly repeated normal shared-DT arm and "off" is connected-zero rotation. Only boundary 839 is an instantaneous counterfactual; later rows are paired longitudinal outcomes.

Boundary	g_{on}	g_{off}	$g_{\text{on}} - g_{\text{off}}$	Ratio	Limit-10 class
839	1.423852	0.565017	0.858835	2.5200	neither
840	1.352574	0.451292	0.901282	2.9971	neither
841	1.605586	0.371823	1.233762	4.3181	neither
842	2.048217	0.338519	1.709698	6.0505	neither
843	1.848073	0.317264	1.530809	5.8250	neither
844	2.508924	0.304979	2.203945	8.2266	neither
845	4.601322	0.301506	4.299816	15.2611	neither
846	4.129866	0.277695	3.852171	14.8720	neither

At boundary 839, enabling rotation therefore raises the pre-clip norm by 0.8588348031044006, or 2.52001478149663×, at this one captured state/input. The A-minus-B global-loss contrast there is −0.00017467141151428223. All eight norm differences are positive, and the seven later signs agree with the first sign. This is persistent longitudinal separation in the observed window, but neither arm crosses 10 at any boundary. It therefore does not establish that rotation caused the historical limit crossing, does not explain the later boundary-853 event, and does not justify a threshold change. The summary remains explicitly non-authorizing.

The retained scheduler record is `diagnostic_runs/matched_intervention_49f06078d19f_1839531_scheduler_record`. Its root-row, full-step, completion-marker, and scheduler-log-hash records have SHA-256 values

```
01713739b41fbc83a04e5a449cdbf702103c3b87568d8087adc1887c178c2053,
74ec9cb245a3086a1ad7afaa0da6479259492b730befaa42f951a1565b5ba672,
5346f5e235063d09bcf8e870fb98a19a05d2e099f99ab0ab8cd5a3735bcda2cd,
23b49df7cf3b72bb6de57273350ce165ce96676015f6e54cf30f70bad4905c38.
```

Scheduler stdout contains exactly one completion marker; its SHA-256 is

```
2760327de1c640203d5bc1552f0807fe298743b5f702997e9b022e51322d38ae.
```

Scheduler stderr is empty. Fresh-node independent validator job `1840069`, submitted from retained runner SHA-256

9e602deac9319a7e12d1d71fbcba75bd6d52da5cdbfcfafd77d34661ff3e7c84,

ran on h200-021-171 from 18:30:14 through 18:39:40 UTC and completed 0:0. It reconstructed both invocation-bound source paths, validated all 4,257 matched-source and 4,253 trigger-source files before and after use, independently returned VALID, rederived the 118-file inventory, and removed all three marker-owned node-local roots before its sole completion marker. Its evidence root is diagnostic_runs/matched_intervention_49f06078d19f_1839531_independent_validation/ evidence-job-1840069. The validation result, acceptance result, and complete 16-row evidence-list SHA-256 values are respectively

55d0fed3dcabcbf9a60a444cce149108f4ec891a428946269a84fcb1330fcbaf,

364d899e1a7e0b999a68917780c562d00638226a6d5640653ecf1b53ad6c3ccf,

19d802d2e8a49f93d9fd8124078f307aa31f6cdf7bb5354bf478757672cb716f.

The retained validator scheduler root-row, all-step, marker, and log-record SHA-256 values are

e186d5a86aa67c75ec0fa070dbd5fb6c78ab3bb947e1d8e51d3ade503bb1c6b1,

5f1e6be98f469a337e6915104b2bb3b02cc8bdb54f711e3108973cd1eb536fe3,

04165dfb52dfe1f9ced189fbb18ed94bdf6b518bc6e28631fa39d8453dbf9451,

974b7bdd010fb454f4063b6b13b705e9f8bdf929032a3a01ab5c4c57c1275cfb.

All retained validator files are mode-0400, single-link regular files and its scheduler stderr is empty.

Remediation work is isolated from the main checkout in

/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_shared_dt_v1/
source_worktrees/rdiag_shared_dt_late_angle_scale_20260904

on branch codex/rdiag-shared-dt-late-angle-scale-20260904. The independently reviewed, non-authorizing design is

docs/plans/2026-09-04-rdiag-shared-dt-late-angle-scale-design-final.md.

It is committed and published with commit, tree, and file SHA-256 respectively

c57c9617f0256189bcc5a733789699bf32731906,

1279bb20cac06d7ee8b5ba231a273cb955b8874e,

549a7592ffe10b3c516f98d023ea67cb00e64ad02c8529df13e2af9325f427ba.

It selects the checkpoint-compatible, rotation-preserving effective angle scale $0.5H^{-1/2} = 0.125$ at $H = 16$, while retaining direct $H^{-1/2} = 0.25$ and connected-zero controls. It does not alter shared_dt_v1. The executable plan is

docs/plans/
2026-09-04-rdiag-shared-dt-late-angle-scale-implementation-plan-final.
md.

It is published with commit, tree, and file SHA-256 respectively

17d86199a0eccf91a4ad8c4e0a256d063efc1d85,

65bd0d9d3c3f3d24b9eb51e4fd42ad327c60e942,

8497caef38ea1fc4811b629704ac97604e3daea8da46830a2865329365ce20e5.

Two formal adversarial plan-review rounds found and drove fixes for exact H=16 oracle coverage, a distinct candidate-above-10 terminal, real-anchor FSDP preflight, complete later-boundary liveness, fail-closed outcome mapping, immutable H200 provenance, prospective scheduler-step counts, and four final CLI/snapshot/validator interfaces. The final plan contains those last repairs and requires their named red/green tests during implementation; the second formal review predated those repairs and therefore remained `NEEDS_REVISION`. Implementation seed commit `68932d5aed62f717b91f8a5c46b38679b70805bb` added only the seven preregistered helper, host, H200, preflight, runner, and validator paths. The reversible H=16 angle-scale override was then implemented and hardened through commits `a14168a...`, `5253faf...`, and `11e7d37...`. Fresh specification and quality reviews accepted that slice after it made rollback best-effort and fail-closed, detected mid-arm scale drift, validated exact per-rank topology evidence, and preserved RNG and state-dict identity. Its focused host contract passes 33 tests; the complete new host file passes 135 tests while nine deliberately red assertions remain for later implementation slices. The matched-prerequisite closure is the current in-progress slice. No numerical or result-bearing GPU job has yet been accepted from this worktree.

The completed prerequisites and diagnostics are the forward localization in job `1836996`, the reviewed shared-DT-only warp-8 RMSNorm-forward control, its passing bounded smoke in job `1837393`, and the passing replacement generation-zero gate in job `1837446`, followed by accepted trigger capture `1838750` and independent validator `1838966`, the valid matched intervention `1839531`, and its fresh-node validator `1840069`. The remaining sequence is:

1. test the narrow, checkpoint-compatible preserve-rotation candidate that changes the effective angle scale from $H^{-1/2}$ to $0.5H^{-1/2}$ against repeated old-scale and connected-zero controls at the same boundary-838 anchor; this test is separately preregistered and non-authorizing;

2. only if that candidate passes its exact repeatability, isolation, liveness, health, and prospective effect-size requirements, freeze its source and rerun all six gates, including an entirely fresh 3300-update soak;

3. build aggregate-v3 only after all six recursive seals validate, then recompute and pin the aggregate, all recipe digests, and all exact execution-manifest digests for that final source; and

4. create the separate required production snapshot and invoke the `prepare-fresh` mode of `submit_rdiag_shared_dt.py` only for `rdiag_kla_shared_dt_v1_750M` with train configuration `tsz128x4k_50B`; retain the supervisor-enforced `standard`, `niah_single_1`, `niah_single_1_8k`, `niah_single_2`, `niah_single_2_8k`, `niah_single_3`, and `niah_multikey_1` evaluation modes.

Job `1835824`'s scientific failure invalidated the prior nine-workload lower bound; jobs `1836996`, `1837446`, `1838750`, and `1839531` have now completed the localization, replacement generation-zero, trigger-capture, and matched-intervention workloads. From this point, even the optimistic path requires at least seven more successful scientific GPU workloads before production launch: one late-anchor candidate probe and the six fresh qualification gates. Production training would be the eighth. Since the demonstrated six-gate control plane expands its six scientific gates to 20 Slurm jobs (six preparation jobs, two H200 control/submit allocations, six gate workloads, and six external attestors), the comparable operational lower bound is at least 24 future scheduler jobs before production submission: one numerical preflight, one actual-anchor FSDP preflight, one candidate allocation, one candidate validator, and the 20-job gate control plane. A held-out replication would increase that count. Failed attempts `1838085` and `1838250` did not satisfy the trigger-capture stage. This count excludes retries, additional discriminator arms, implementation-time GPU checks, held-out causal replication, and production requeues. Until that sequence closes, neither a production authorization nor a replacement training result is claimed.

A Reproducibility audit

RDiag implementation baseline. The sealed E2 baseline implementation statements in this note were audited against branch `rdiag` at Git commit `c3346feaecae1486cdbc71f82b4edeacc4198950` (tree `88e423d2926d8ae81945c08393a59861d285907c`). This pin identifies the sealed implementation snapshot; the present documentation-only working-tree update is intentionally outside that snapshot. Relevant regions are:

- `lit_gpt/rdiag_kla.py`, construction and buffers around lines 137–193, phase construction around 222–293, and the fused layer route around 409–516;
- `lit_gpt/kla_ops/rdiag_clock.py`, anchor and eager oracle around 72–119 and 190–270, exact fused clock around 283–390, and native-order custom backward beginning around line 515;
- `lit_gpt/kla_ops/rdiag_direct.py`, the unchanged direct-pair forward around line 162, guarded direct-pair backward around line 338, precise clock materializer around line 300, and radial wrapper around line 920;
- `lit_gpt/kla_ops/rdiag_phase.py`, negative scan contract and reference around 1–19 and 294–328;
- `lit_gpt/kla_ops/rdiag_bounded_phase.py`, bounded fail-visible phase contract around 1–11 and 41–172;
- `lit_gpt/kla_ops/rdiag_norm.py`, public fused normalization/rotation contract around 10–90;
- `lit_gpt/kla_ops/rdiag_norm_reference.py`, normalization, BF16 boundary, and adjacent-pair rotation around 24–96;
- `lit_gpt/kla_ops/diag_naive.py`, KLA equations around 14–32, 144–199, and 451–491;
- `lit_gpt/diag_kla.py`, process-noise and isotropic-output initialization around 280–300;
- `lit_gpt/config.py`, bounded RDiag recipe fields around 349–369 and 742–763;
- `lit_gpt/rdiag_training_health.py`, health thresholds and observed counters around 24–37 and 121–255, and fail-closed decision logic around 579–612;
- `lit_gpt/fsdp_strategy.py`, the FP32-buffer mixed-precision override beginning around line 131;
- `scripts/tests/rdiag/test_rdiag_bounded_h200.py`, exact radial, actual-logit, exceptional-value, full-layer parity, and A/B performance gates;
- `sanboxes/kernels/rotational_diag_kla_v1/DESIGN.md`, moving-frame semantics and the pair-tied literal model; and
- `sanboxes/kernels/rotational_diag_kla_v1/RESULTS.md`, the correctness and v1 performance evidence around lines 128–166.

The whole tree is sealed by the snapshot manifest identified below, which is the authoritative byte-level inventory. Three central files in the final parity implementation and gate have SHA-256 values

- `lit_gpt/kla_ops/rdiag_clock.py`,
  `d02a3f0d0994067afc174f1a77f7ebd8fcecd73c3a18956e72ab79ab49afc966`;
- `lit_gpt/kla_ops/rdiag_direct.py`,
  `9fa3e29102bfa512ecbc57fc04ce6453e3a33e7aed61e7049026b71fe81faace`;
- `scripts/tests/rdiag/test_rdiag_bounded_h200.py`,
  `68d7db9b6793a33ef2574c7a39527306c3b3b500a03e1807790ce59889be8517`.

The final result followed three deliberately narrow diagnoses. Commit `fe1b1258` made the radial clock mean bitwise exact but left three full-layer angle-weight mismatches in job 1777082. Job 1777229 then proved that the first remaining forward divergence occurred at sigmoid output despite a bitwise-identical input logit. Commit `4ce6e19` matched the CUDA-eager sigmoid and angle VJP, and commit `c3346fe` added the missing-delta compile-time guard that makes the detached radial path immune to nonfinite discarded angles.

Installed Mamba implementation. The source audit used `mamba-ssm 2.3.2.post1` installed under the active `hdn` environment; those bytes match upstream tag `v2.3.2.post1` at commit `a14b1dff0454a3bc27d9eb31355dc01e4b2490ec`. Paths and SHA-256 identities are:

- `mamba_ssm/modules/mamba2.py`,
  `605e4439ff0baec8d8acaf4a191d9f0570eea9900065a065909124c472b08707`; inspect roughly lines 95–136, 210–260, and 309–318;
- `mamba_ssm/ops/triton/ssd_combined.py`,
  `39c19c1e4c8e36982847079bc12b4f07ed3af03ac68f6a6653200ef80df5515d`; this is the fused Mamba-2 SSD scan;
- `mamba_ssm/ops/triton/ssd_chunk_state.py`,
  `6d82771b2f62a7bf84c381c8ef8e4b579a95e0352021d117507a1072e4d71dcd`; inspect roughly lines 40–86 and 102–174;
- `mamba_ssm/modules/ssd_minimal.py`,
  `6442a9fe25a6792258df657392ab1fab4dd873ee777d9dd5783cc76030218e07`; this is the readable Mamba-2 SSD reference;
- `mamba_ssm/modules/mamba3.py`,
  `de1b104865c941eb6eeac63c3a75d5913fc2e7a119d52991590ef19c381b05ac`; inspect roughly lines 76–87, 150–178, and 220–237;
- `mamba_ssm/ops/triton/mamba3/angle_dt.py`,
  `7d99f8cff37371cfbfe9d4dd36f17a16135fb07f2c6e14add3ad4dd5c0034e2e`; inspect roughly lines 83–118 and 298–337;
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_combined.py`,
  `410ad57555b5841fba7bd76fdafdc3ddf82741176489c3e3493f5c6f3c0c6012`; inspect roughly lines 90–105 and 247–259;
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_fwd.py`,
  `68d8d0dba0e3913f2e996862603289fd0d7157e7dc57c872e04234be8bb82695`; inspect roughly lines 256–332, 348–352, and 386–425; and
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_bwd.py`,
  `aa1b0aa8f8f7950503adf55220d3feebfea2fe006dc418b5a4e75e1e6842eadc`; inspect roughly lines 978–1051 and 1475–1550.

The hashes, package version, paths, and upstream commit are the durable identity; line numbers are navigation hints. Repository Mamba-3 SISO dispatch goes through the FLA wrapper. Its executable baseline uses the patched `hdn_mamba` environment; the local compatibility/decay patches do not alter the angle scan or adjacent-pair rotation formulas. The unpatched `hdn` FLA wrapper is not the runnable decay-correct baseline and was not used to infer those formulas.

Qualification evidence and present status. The lower-tail rejection is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/probe_v1_1774423/probe_trace.json`. The redesigned E2 decision is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/qualification_bounded_de_20260828_1145/D_E_DECISION.json`. The precision experiment is recorded in `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/radial_clock_precision_1776573/result.json`; focused VJP evidence is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_vjp_diag_k64_1776509.log`, and the five-case composed check is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_angle_order_1776788.log`. These artifacts support the specific debugging observations in Section 8.

The sealed implementation snapshot used by the one-H200 gate is `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/snapshots/production_e2_20260828_2344`. Its `SOURCE_MANIFEST.json` SHA-256 is

`05db76db18b8f4fe170a102e2e5df28bb54a4e36ca7ebe4565a1d813b44530a3.`

The authoritative qualification log is `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_h200_1777777.log`. SLURM job 1777777 completed with status `COMPLETED`, exit code `0:0`, 117 passing core tests, and 54 passing E2 tests. This closes the one-H200 kernel/layer gate. Distributed training-system checks and the multi-node soak/attestation remain before submission; training completion and the full post-training evaluation are subsequent empirical deliverables.